No Regrets in clustered wind farms: how wind rose directionality and optimization thoroughness shape the cost of ignoring neighbors

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Abstract. We study *design regret* in clustered wind farms: the AEP gap between a layout optimized with knowledge of neighboring-farm wakes and one optimized in isolation, evaluated under the same neighbors. Our main contribution is methodological. Computing regret requires differencing two large, noisy AEP-optimization outcomes; this catastrophic cancellation amplifies optimizer noise into the difference. We show that regret estimates do not stabilize until $K = 500$ multistart restarts—one to two orders of magnitude beyond standard practice—that convergence in K depends on the size of the regret signal relative to multistart noise, and that the estimate is conditional on the inner optimizer: a stronger iteration schedule raises the regret found at every restart budget, so any inter-farm coordination study computing AEP differences between strategies inherits these requirements. With convergence quantified by exact subsampling of stored per-start outcomes, we map design regret across the elliptical wind rose space of Hart (2025). We bound regret two ways: greedy adversarial individual-turbine placement (upper bound, dominated by wind rose folding) and regret cross-sections with an identical neighboring farm (realistic scenario, representative of typical lease layouts). Both quantities span an order of magnitude across the two engineering wake models we test: under Bastankhah, adversarial regret is 0.4–2.5 % of AEP and realistic 0.16–0.49 %; under TurboPark, realistic regret alone reaches 1.9–10.6 %, overtaking the adversarial bound computed with a weaker optimizer—under-optimization can masquerade as safety. The adversarial figure is a deliberate ceiling—it assumes a neighbor placing turbines purely to damage you—while the identical-copy figure is a specific, plausible scenario rather than a proven bound: a self-interested neighbor optimizing its own production is not covered by either, and we flag it as the main open case. The two wake models also disagree on where vulnerability peaks: oblique to the prevailing wind under Bastankhah's fast recovery, at or near directly upwind under TurboPark—a contrast that is unambiguous in a single-direction stress test (135° vs. 90° inflow). For policy, we invert the buffer-decay curves into a wind-rose-conditional buffer lookup table. Under Bastankhah, capping realistic-scenario regret at 0.2% of AEP requires at most $\sim$8–9D ($\sim$2 km) for typical North Sea sites—an upper bound, since two conservative post-processing choices each inflate it and removing both gives $\sim$4–6D. Under TurboPark—which recovers wakes more slowly and pushes realistic regret an order of magnitude higher—the equivalent 2% threshold is never reached anywhere within the swept range at any of those sites, so the requirement there is censored at $> 40D$ (> 9.6 km), already beyond common spacing recommendations. Finally, ringing the target with up to eight neighboring farms shows that regret is non-monotonic in the number of neighbors: it decomposes into AEP loss times the fraction a redesign can recover, and adding farms raises the first

while lowering the second. Lease geometry caps the problem—farms $\sim 150D$ across admit at most four close neighbors, so the many-close-neighbors case is unbuildable rather than unstudied—and a controlled experiment holding neighbor capacity fixed shows that what closes off redesign is concentrated wake mass in the energy-carrying directions rather than angular symmetry. The 12–22× wake-model bracket dominates any single number a policy rule could quote; rather than picking one, we tabulate both and recommend buffer policies treat the bracket explicitly.

1 Introduction

The North Sea is entering an era of unprecedented wind farm clustering. Denmark's Vindmølleparken energy island will connect up to 10 GW across multiple farms in a shared concession zone; the German Bight already hosts over 8 GW with plans for 30 GW by 2050; and the broader North Sea basin is projected to reach 300 GW as part of Europe's 500 GW offshore target. At this density, wind farms no longer operate in isolation. Cluster wake losses in the German Bight are projected to reduce capacity factors by 30 %, with half of the loss attributable to cross-border wake accumulation (Fliegner et al., 2025). The phenomenon—increasingly termed "wind theft" (Finseraas et al., 2024; Du et al., 2026)—has prompted the UK to issue the first national policy guidance on inter-farm wake effects and NYSERDA to recommend minimum inter-farm spacing of 4 nmi (NYSERDA, 2018). Wake-induced velocity deficits have been measured at distances exceeding 50 km (Schneemann et al., 2020; Canadillas et al., 2022), and inter-farm losses are projected to exceed the impact of a century of climate change on North Sea wind resources (Warder and Piggott, 2025).

Yet these emerging policy responses share a common gap: they treat all sites uniformly. Buffer distance recommendations, "good neighbour" design principles, and compensation frameworks do not account for variation in wind climate. This is a significant omission. A site where wind blows predominantly from one direction creates a single, predictable wake corridor that a neighbor-aware layout can avoid. A site with bidirectional wind creates competing corridors that no single rearrangement can escape. The value of coordinated design—and the cost of failing to coordinate—should depend fundamentally on the directional structure of the wind resource.

We formalize this intuition through the concept of *design regret*: the AEP difference between a layout designed with perfect neighbor foresight and one designed in isolation, both evaluated under the same neighbor wakes:

$$R = \mathrm{AEP}(\mathbf{x}^*_{\mathrm{cons}}, \mathbf{n}) - \mathrm{AEP}(\mathbf{x}^*_{\mathrm{lib}}, \mathbf{n}). \tag{1}$$

Design regret isolates the cost of the design decision itself, separate from the total AEP reduction caused by neighbor wakes. By sweeping the elliptical wind rose parameterization of Hart (2025)—which separates directional concentration from bidirectionality—we quantify how wind rose shape governs design regret. The answer is dramatic: under Bastankhah, adversarial regret varies six-fold across the shape space; under TurboPark, even a realistic identical-neighbor scenario spans 2.0–10.6 % of AEP across the same shape sweep. Buffer distance requirements inherit this dependence.

Quick et al. (2026) applied this framework to the Danish Energy Island cluster and found negligible design regret for all nine neighbor configurations. We show this is not a general property of offshore wind, but a consequence of DEI's partially bidirectional wind climate ($f \approx 0.38$). Sites in the southern North Sea with strongly unidirectional southwesterly winds sit in the high-regret region of the shape space and are substantially more vulnerable to uncoordinated cluster development.

However, accurately computing these quantities reveals a methodological challenge that undermines all existing regret-like calculations. Layout optimization is highly multimodal (Thomas et al., 2023; Valotta Rodrigues et al., 2024), and regret—as the *difference* between two optimization outcomes—amplifies the effect of local optima. We show that at least 500 restarts are needed for the regret estimate to stabilize—over an order of magnitude more than typical practice.

This paper makes five contributions:

1. A convergence study demonstrating that hundreds of multistart optimizations are needed for reliable regret estimation—over an order of magnitude more than typical practice—and that the required budget depends on both the size of the regret signal and the strength of the inner optimizer.

2. A systematic sweep of the wind rose shape space showing that design regret varies several-fold with directionality, with concentrated unidirectional sites most vulnerable under both wake models tested.

3. Buffer distance analysis quantifying how neighbor separation reduces regret, how the decay rate depends on wind rose shape, and how the absolute buffer requirement depends on the wake model ($\sim$2 km under Bastankhah, versus a requirement still unmet at 9.6 km under TurboPark, for the same real sites).

4. Regret cross-sections with a realistic identical neighbor farm under two wake models, showing that the adversarial–realistic gap and even the location of peak vulnerability (oblique vs. near-upwind) are wake-model-dependent, and that a realistic scenario computed with a stronger optimizer can exceed an adversarial bound computed with a weaker one.

5. A ring sweep over the number of neighboring farms, showing that regret is non-monotonic in neighbor count, that lease geometry makes the many-close-neighbors case unbuildable, and—via a controlled experiment holding neighbor capacity and distance fixed—that concentrated wake mass rather than angular symmetry governs how much damage a redesign can recover.

2 Methods

2.1 Problem formulation

We consider a target wind farm of $N_t = 50$ IEA 15 MW turbines ($D = 240\,m$, hub height 150 m) within a fixed polygonal boundary derived from the Danish Energy Island concession zone. Minimum inter-turbine spacing is $4D$ (960 m). This configuration is representative of near-future North Sea developments: the IEA 15 MW turbine is the reference class for planned GW-scale projects, and the DEI polygon provides a realistic, irregular boundary.

The liberal layout $\mathbf{x}^*_{\text{lib}}$ maximizes AEP assuming no external turbines. The conservative layout $\mathbf{x}^*_{\text{cons}}$ maximizes AEP with specified neighbors $\mathbf{n}$. Both are evaluated with neighbors present to compute regret via Eq. (1). By construction, $R \geq 0$ when both optimizations reach a global optimum, since the conservative search has access to all of the liberal layout's design space plus the additional information about neighbors. With finite multistart budgets neither optimization is guaranteed to be globally optimal, and stochastic noise can occasionally produce $R < 0$. We therefore include the liberal layout itself as one of the candidate starts in the conservative search and define

$$R = \max(\text{best conservative AEP}, \text{ AEP of liberal layout with neighbors}) - \text{AEP of liberal layout with neighbors}. \quad (2)$$

This *regret pooling* step is a one-sided clip, and it is upward-biased whenever it binds: it replaces a negative measurement with zero but never the reverse. On the Bastankhah grid the clip binds in 1102 of 5880 evaluations (18.7 %) and raises the mean regret from 2.24 to 2.44 GWh, an inflation of 8.8 %; under TurboPark, whose signal is far above the noise, it binds rarely and the inflation is negligible. We retain the clip because a negative value is not physically meaningful—the conservative search has access to the liberal layout's design space—but the frequency with which it binds is itself a diagnostic that the underlying search has not resolved the conservative optimum, and we report it rather than treating the pooled value as unbiased. The same trick is needed for any AEP-difference study where $R \geq 0$ holds in theory but optimization is imperfect.

2.2 Wake model

We use the Bastankhah and Porté-Agel (2014) Gaussian deficit model with $k = 0.04$ and root-sum-of-squares superposition. Our implementation applies a radial cut-off, zeroing the Gaussian deficit beyond twice its standard deviation. This is consequential for the Bastankhah results and we treat it as a source of systematic bias rather than a modelling choice: it inflates neighbour-induced AEP loss on the headline configuration by 18% under Bastankhah and 5% under TurboPark, and because regret is a small difference between two optimised AEPs the effect is amplified in the Bastankhah regret figures. The TurboPark results, which carry the paper's central quantitative claim, are insensitive to it. The simulation is implemented in JAX (Bradbury et al., 2018), enabling automatic differentiation through the wake model. AEP is computed over 24 equally-spaced wind direction sectors. The elliptical-rose cases use a constant 9 m/s; the DEI case instead uses the per-sector mean speed from the measured record, which spans 7.5–11.6 m/s (mean 10.1 m/s). Both therefore evaluate $P(\mathbb{E}[v])$ rather than $\mathbb{E}[P(v)]$, which is discussed as a limitation in Section 4.4:

$$\text{AEP} = 8760 \sum_{k=1}^{24} w_k \sum_{i=1}^{N_t} P_i(U_{i,k}), \quad (3)$$

where w_k is the sector frequency, P_i is the turbine power curve, and $U_{i,k}$ is the effective wind speed at turbine i under direction k.

2.3 Wind rose parameterization

Wind rose shape is parameterized using the elliptical model of Hart (2025), which derives the directional frequency distribution from a unit-area ellipse with semi-major axis a (along the prevailing-direction axis) and semi-minor axis $b = 1/(\pi a)$ (perpendicular). The CDF of a direction θ relative to the prevailing axis is built from the first-quadrant expression (Hart 2025, Eq. 2):

$$F_{Q1}(\theta) = \frac{1}{2\pi} \arctan\left(\pi a^2 \tan\theta\right), \qquad \theta \in [0, \pi/2), \quad (4)$$

with $F_{Q1}(\pi/2) = 1/4$, and extended to the full circle via four-fold symmetry. Sector probabilities are obtained by differencing the CDF at the sector edges, rotating by the prevailing direction θ_{prev}, and applying a folding multiplier that breaks the front/back symmetry (Hart 2025, Eqs. 3–5):

$$p_i = \underbrace{\left[\tilde{F}(\theta_i + \Delta) - \tilde{F}(\theta_i - \Delta)\right]}_{\text{un-folded elliptical probability}} \cdot \underbrace{\begin{cases} 1+f, & \theta_i - \theta_{\text{prev}} \in (-\pi/2, \pi/2) \quad \text{(front half)} \\ 1-f, & \theta_i - \theta_{\text{prev}} \in (\pi/2, 3\pi/2) \quad \text{(back half)} \\ 1, & \theta_i - \theta_{\text{prev}} = \pm\pi/2 \quad \text{(perpendicular to prevailing)} \end{cases}}_{\text{folding multiplier}}, \tag{5}$$

where θ_i is the sector centre, $\Delta = \pi/n_{\text{sec}}$ is the half-sector width, and $\tilde{F}$ is the rotated, periodically-extended CDF $\tilde{F}(\theta) = F(\theta - \theta_{\text{prev}}) + \lfloor(\theta - \theta_{\text{prev}})/(2\pi)\rfloor$. Two shape parameters control the distribution:

- **Shape parameter** $a > 0$: controls the ellipse eccentricity. The special value $a = 1/\sqrt{\pi} \approx 0.564$ gives a circle (uniform distribution). For $a > 1/\sqrt{\pi}$ the ellipse is elongated along the prevailing axis and probability concentrates there (larger $a \Rightarrow$ more concentrated unidirectional rose when $f = 1$). For $a < 1/\sqrt{\pi}$ the ellipse is elongated perpendicular to the prevailing axis, producing a bimodal rose with peaks at $\theta_{\text{prev}} \pm 90°$.

- **Folding parameter** $f \in [0, 1]$: controls front/back symmetry. At $f = 0$ the rose is bidirectional (equal probability from θ_{prev} and $\theta_{\text{prev}} + 180°$); at $f = 1$ all back-half probability is folded onto the front half.

This parameterization separates two physically distinct wind climate properties: how concentrated the energy is along an axis (controlled by a) and whether the rose is symmetric about its prevailing direction (controlled by f). The prevailing direction is fixed at $\theta_{\text{prev}} = 270°$ and wind speed at 9 m/s across 24 equally-spaced sectors. We sweep $a \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ and $f \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (35 combinations). This range spans the variety of North Sea wind climates, from the strongly unidirectional southwesterlies of the German Bight to the more bidirectional conditions at the Danish Energy Island.

Fitting the DEI wind rose with the elliptical model yields $a \approx 0.64$, $f \approx 0.38$ ($R^2 = 0.84$), placing it in the moderate-concentration, partially-bidirectional regime.

2.4 Adversarial neighbor placement: upper bound on regret

To establish a worst-case upper bound on design regret, we place individual neighbor turbines using a greedy grid search. A candidate grid with $5D$ spacing extends beyond the target boundary, minus a $2D$ exclusion buffer. The extent differs between the two production sweeps ($50D$ for Bastankhah, giving 1567 candidates; $200D$ for TurboPark, giving 10373), which matters: enlarging the grid admits more distant, less damaging candidates and so is not neutral (Appendix E). At each of 30 greedy steps, the single candidate maximizing incremental regret is selected. Evaluating regret at every candidate would require a full K-start conservative optimization per candidate (thousands of SGD solves per step); instead we use a two-stage screen-then-evaluate scheme. *Stage 1 (cheap screening):* for every remaining candidate, we compute the AEP loss that placing a turbine there would induce on the *liberal* layout—a single forward wake simulation, no inner optimization. *Stage 2 (full evaluation):*

the top few candidates by screening AEP loss receive full K-start conservative re-optimization; the one producing the highest regret is selected and placed. Production runs used the top 5 candidates for the elliptical-rose sweeps and 8–10 for the DEI case. This reduces per-step cost by a factor of 20-25 at the same final result, exploiting the strong correlation between raw AEP loss (which any upwind position induces) and the eventual regret after re-optimization. Figure 1 shows a mid-greedy snapshot, the screening AEP-loss field over the candidate grid, and the candidates that would be fully evaluated at the next step.

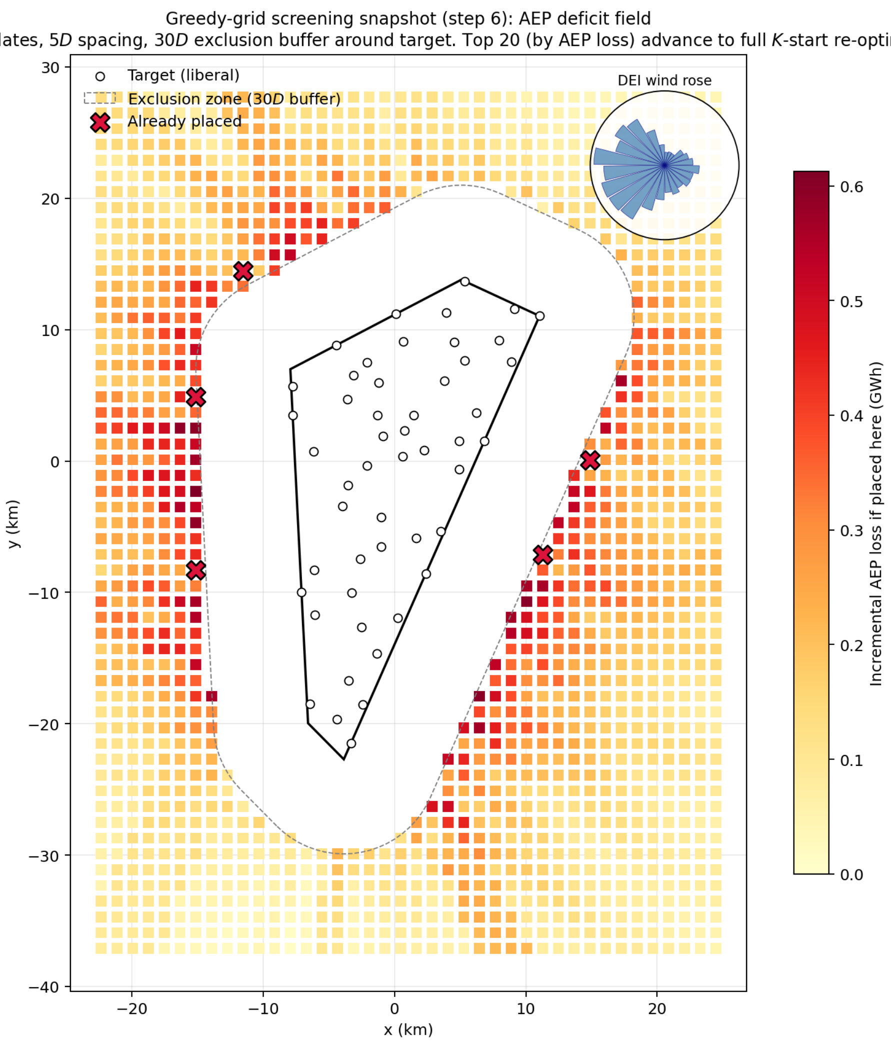

Figure 1. Greedy-grid screening snapshot at step 11/30 (10 neighbors already placed). Colored squares: candidate grid (spacing $5D$, extent $30D$ beyond target boundary), colored by the incremental AEP loss that placing a neighbor at that grid point would induce on the liberal target layout (single forward sim, no optimization). Red X markers: neighbors already placed in prior steps. Only the top few candidates by screening AEP loss advance to full K-start re-optimization in Stage 2 (5 in the production sweeps). Dashed gray: $2D$ exclusion zone around the target polygon. White circles: target liberal layout.

This adversarial approach places unconstrained individual turbines—not realistic wind farms—and is designed to find the *maximum achievable regret* for a given wind rose. It provides an upper bound because: (i) individual turbines operate at full thrust coefficient, whereas turbines within a real neighboring farm would experience internal wake losses; and (ii) the greedy algorithm optimizes placement purely to damage the target, whereas a real developer would optimize for their own production. The results from this analysis should be interpreted as the ceiling on design regret, not as a prediction of what a realistic neighbor would cause.

2.5 Regret cross-sections: realistic neighbor scenario

To assess design regret under more realistic conditions, we introduce the *regret cross-section*. An identical copy of the target farm—with the same number of turbines, boundary polygon, and liberal-optimized layout ($K_{\text{lib}} = K_{\text{cons}}$; see Sect. 2.6)—is placed at each combination of 24 bearings (15° steps) and 7 buffer distances (2, 5, 10, 15, 20, 30, 40D), producing 168 evaluations per wind rose case (Fig. 2).

2.5.1 Inter-farm distance definition.

Buffer distance is defined as the minimum Euclidean distance between the boundary polygons of the target and reference farms (not centroid-to-centroid). For each (bearing, distance) pair, we iteratively compute the centroid offset required to achieve the specified boundary gap, accounting for the irregular polygon shape. At each configuration, we also record the minimum turbine-to-turbine distance between the two farms. This boundary-gap definition ensures that no turbines from the reference farm overlap the target farm area at any bearing or distance, which is critical for physically meaningful results.

2.5.2 Reference farm construction.

The reference farm is an identical translated copy of the target: same boundary polygon, same number of turbines ($N_t = 50$), and same liberal-optimized layout. This represents the most natural "what-if" scenario: a neighboring developer builds the same farm design nearby.

At each position, the target farm is re-optimized with K_{cons} restarts (matching K_{lib}) while the reference farm remains fixed. The resulting polar map separates bearing and distance effects and reveals whether vulnerability is isotropic or directionally concentrated.

2.6 Multistart optimization and convergence

Each layout optimization uses gradient-based stochastic gradient descent (SGD) (Quick et al., 2023). Two iteration schedules are used. The greedy grid sweep uses a baseline schedule: a constant learning-rate phase of 5000 steps followed by a 5000-step decay phase, i.e. 10000 gradient steps in total. All cross-section and buffer-table production runs use a tuned learning-rate schedule (referred to as the *FunWake* schedule below; 5000 iterations), obtained by an automated schedule search on the liberal layout problem. Note that the two schedules therefore do *not* run the same number of gradient steps: the baseline takes 10000

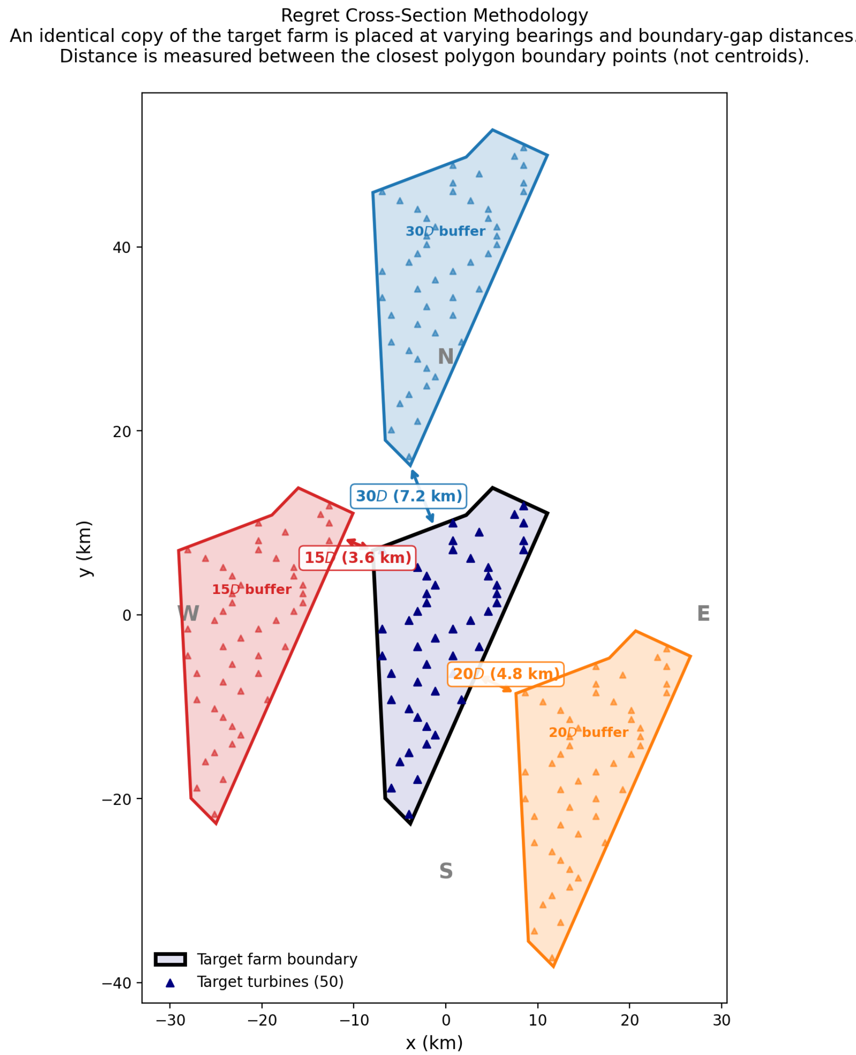

Figure 2. Regret cross-section methodology. Left: the target farm (gray polygon with blue turbines) surrounded by example reference farm placements at four bearings. Buffer distance is measured between polygon boundaries, not centroids. Right: the resulting polar regret map, with wind rose inset.

and FunWake 5000, so FunWake reaches higher-AEP optima at half the step count rather than at parity. This matters for regret because an under-optimized inner solver mis-estimates the conservative optimum—and, as Section 3.7 shows by holding a neighbour configuration fixed and varying only the schedule, that mis-estimation is worth a factor of 1.1–2.1 in the reported number.

Restarts include grid initialization, the liberal layout (ensuring regret pooling), and random polygon-sampled positions. The liberal layout is included as one of the conservative *restarts*; because the solver returns its final iterate rather than the best iterate seen, this does not by itself guarantee $R \geq 0$, and non-negativity is imposed by the explicit clip in Eq. (2). Following the convergence analysis of Quick et al. (2026), who found ∼400 starts needed for 66-turbine liberal layout convergence, we perform a dedicated convergence study for the regret computation (Section 3.1).

Production configurations: $K = 500$ with the baseline schedule for the greedy grid sweep (upper bound); $K_{\text{lib}} = K_{\text{cons}} = 2000$ with the FunWake schedule for the elliptical-rose cross-sections, the buffer tables, and the near-distance DEI cells; and $K_{\text{lib}} = K_{\text{cons}} = 500$ with the FunWake schedule for the far-distance DEI cells and the single-direction case. The $K = 500$ budget is justified by a per-start subsampling analysis showing that it captures large regret signals (TurboPark, single-direction) to within a few percent; the corresponding small-signal Bastankhah cells carry the same lower-bound caveat as the rest of the Bastankhah grid (Section 3.1). Equal K is always used for liberal and conservative optimization to avoid asymmetry artifacts.

3 Results

3.1 Convergence study: how many starts are enough?

We sweep K from 1 to 2000 for three configurations spanning the wind rose space (Table 1, Fig. 3). A 5×5 reference neighbor farm is placed at a fixed bearing and distance for each.

Table 1. Design regret (GWh/yr) at selected K values.

	Design regret (GWh/yr) at start budget K					
Configuration	100	300	500	2000	3000	5000
Bastankhah						
Conc. unidir. ($5D$)	15.9	26.2	26.2	26.6	26.6	27.4
Conc. unidir. ($20D$)	67.9	79.3	79.3	79.3	79.3	79.3
Mod. bidir. ($5D$)	43.1	43.6	43.6	43.6	**52.6**	52.6
TurboPark						
Conc. unidir. ($5D$)	72.1	72.1	75.7	**111.1**	111.1	111.1
Conc. unidir. ($20D$)	30.1	45.9	54.3	54.3	54.3	55.8
Mod. bidir. ($5D$)	57.7	57.7	57.7	60.5	60.5	**67.2**

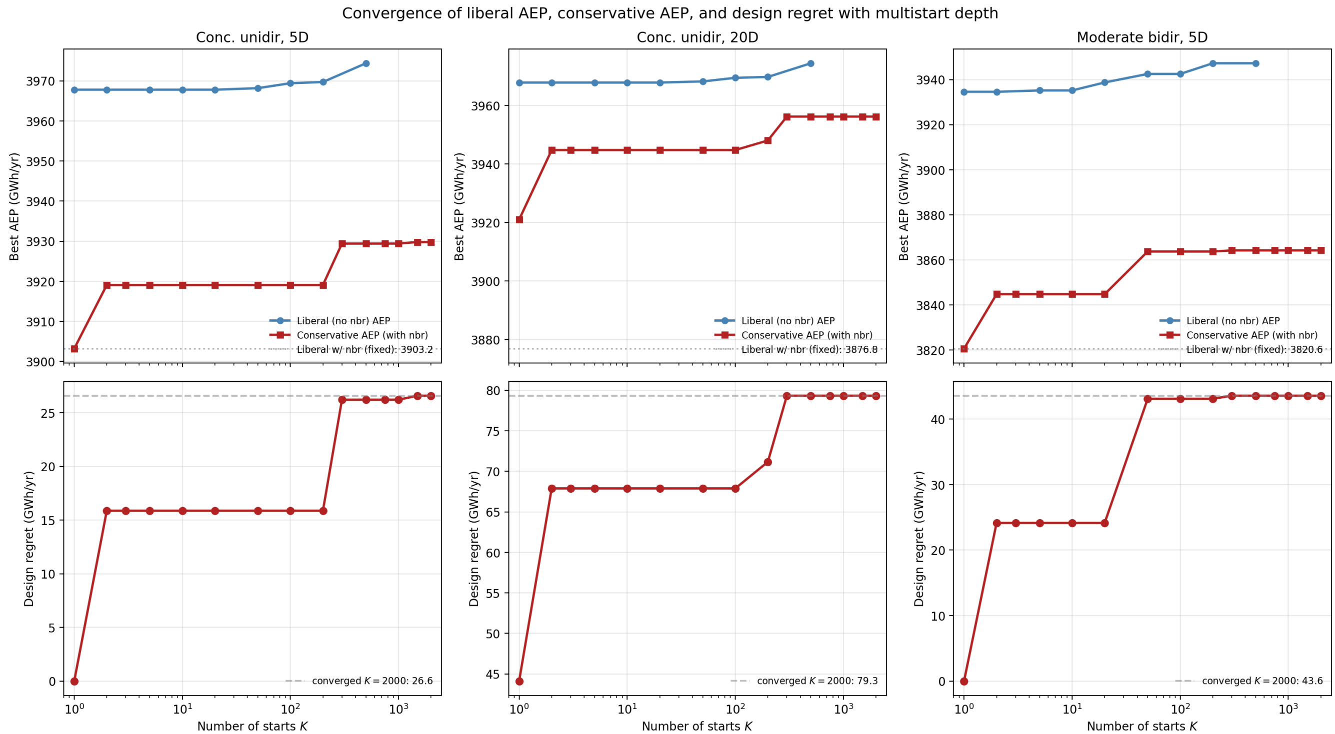

Figure 3. Convergence of design regret with multistart depth K (baseline schedule). Dashed gray: converged value ($K = 2000$). Most of the gain is realised by $K \approx 500$, but four of the six configurations step up again beyond it (Table 1); see Fig. 4 for the schedule- and signal-size-dependence of this statement.

At $K = 100$, the concentrated unidirectional case at $5D$ still underestimates regret by $\sim$40 %, because the landscape has many local optima with similar standalone AEP but different neighbor-present AEP. Most of the gain is realised by $K = 500$, but—and this is the more important observation—*none* of these curves is flat thereafter. Four of the six configurations take a further discrete step upward beyond $K = 500$: the Bastankhah moderate-bidirectional case jumps 21 % at $K = 3000$, and the TurboPark concentrated-unidirectional case jumps 47 % at $K = 2000$ and a further 11 % by $K = 5000$ on the far-buffer case. Because the estimator is a running maximum over starts it can only increase with budget, so every value in Table 1 is a lower bound, and we have not established an upper one. We therefore do not claim convergence at any budget tested; we claim that $K \lesssim 100$ is indefensible and that reported magnitudes should be read as lower bounds that grow slowly with budget. SGD iteration convergence is better behaved: at $K = 500$, regret at 5000 iterations is within 3 % of the 10000-iteration value.

The root cause is catastrophic cancellation: regret is a *difference* between two large AEP values, so a 0.5 % error in either optimization can produce a 50 % error in regret. This applies to any study computing AEP differences between strategies—robust optimization, Pareto analysis, or inter-farm coordination assessments.

3.1.1 Per-start subsampling analysis.

The production cross-section runs store the conservative AEP of every individual start, which permits an exact convergence analysis without re-running anything: for each evaluation, the expected regret at a budget of K starts is computed from the order statistics of a size-K subsample drawn without replacement from the 2000 stored starts. Figure 4 shows the result across all 245 (a, f, d) grid cells per wake model. Two findings sharpen the convergence picture. First, convergence depends on the size of the regret signal relative to start-to-start noise: under TurboPark, whose regret signal is large, $K = 500$ captures a median 96 % of the $K = 2000$ regret (5th percentile 89 %); under Bastankhah, whose realistic-scenario regret sits near the noise floor, the median capture at $K = 500$ is only 54 % and the expected regret is still rising at $K = 2000$. Second, convergence in K is conditional on the inner optimizer: these curves are computed with the tuned FunWake schedule, and a weaker schedule shifts the entire curve downward rather than converging to the same value. Reported Bastankhah regret magnitudes should therefore be read as lower bounds, while the TurboPark magnitudes are converged to within a few percent. This analysis also justifies running the supplemental far-distance DEI and single-direction cells at $K = 500$: their TurboPark and single-direction regret signals are large (Section 3.5), placing them in the well-captured regime, while their Bastankhah values carry the same lower-bound caveat as the rest of the Bastankhah grid.

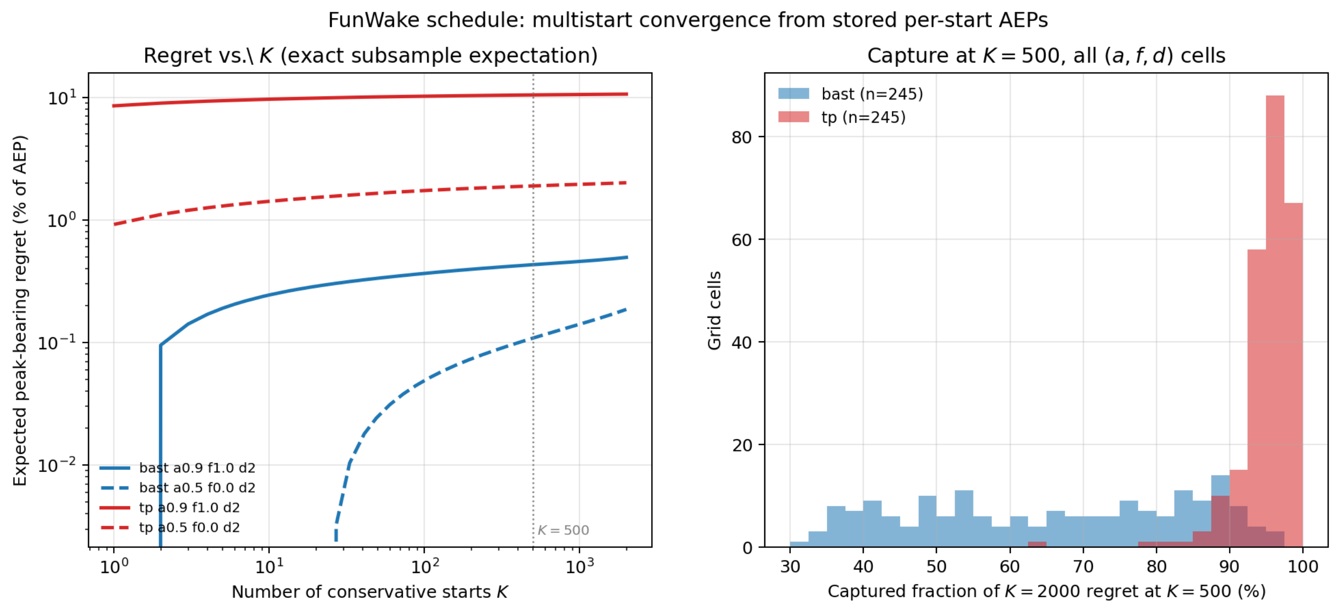

Figure 4. Multistart convergence computed exactly from stored per-start AEPs (FunWake schedule). Left: expected peak-bearing regret vs. start budget K for representative high- and low-regret cells (exact subsample expectation; no bootstrap noise). Right: fraction of the $K = 2000$ regret captured at $K = 500$ across all 245 buffer-table cells per wake model. TurboPark (red) is converged to within a few percent at $K = 500$; Bastankhah (blue), whose regret signal sits near the multistart noise floor, is not converged even at $K = 2000$.

3.2 Wind rose shape controls design regret

Figure 5 shows design regret from the greedy adversarial sweep across the (a, f) space at $K = 500$ (baseline schedule, as is the buffer-decay analysis of Section 3.4; the buffer tables of Section 3.3 and the cross-sections of Sections 3.5–3.8 use the converged FunWake configuration). Under Bastankhah, regret ranges from 17.2 GWh (0.4 % of AEP) at $a = 0.5$, $f = 0.0$ to 100.6 GWh (2.5 %) at $a = 0.9$, $f = 1.0$—a six-fold range. Under TurboPark, the same sweep yields 2.6 % to 9.0 % of AEP across (a, f), with the same qualitative ordering (folding dominant, peak at $a = 0.9$, $f = 1$) but 3.5–6.5× higher magnitudes.

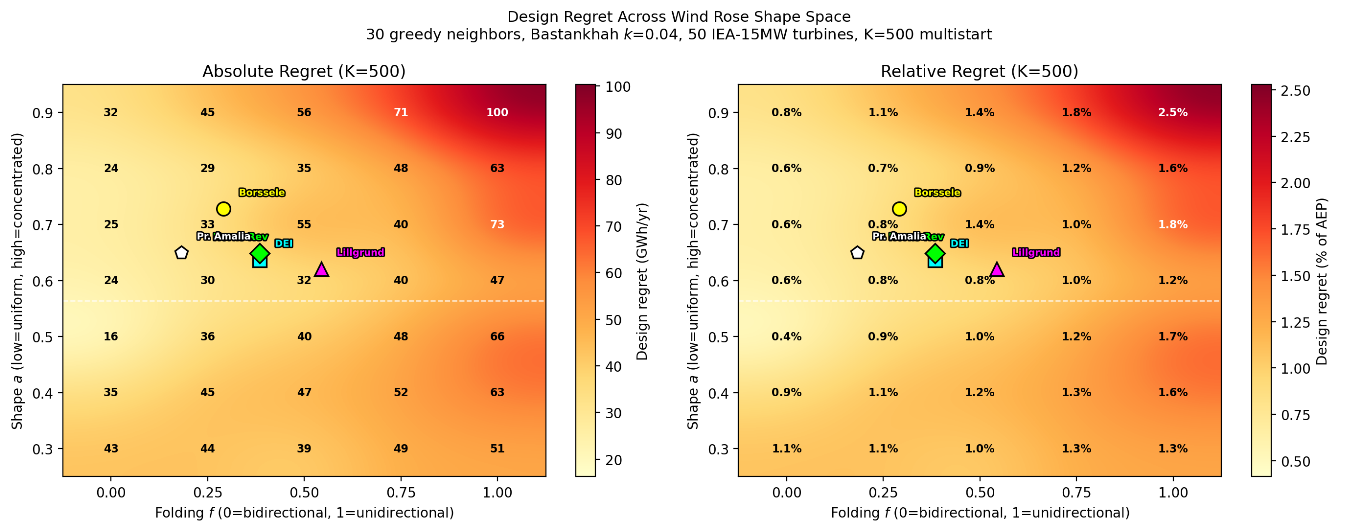

Figure 5. Design regret across the wind rose shape space at $K = 500$. Left: absolute (GWh/yr). Right: relative (% of AEP). Markers show fitted parameters for real North Sea sites: DEI (cyan square), Horns Rev 1 (green diamond), Lillgrund (magenta triangle), Borssele (yellow circle), and Princess Amalia (white pentagon). All real sites fall in the moderate-regret region of the shape space.

The folding parameter f is the dominant factor (Fig. 6). Mean regret increases monotonically from 45.2 GWh at $f = 0$ to 77.9 GWh at $f = 1$. Concentration (a) has a weaker, non-monotonic effect.

Unidirectional wind ($f = 1$) creates a single wake corridor that the liberal layout is blind to; the conservative optimizer shifts turbines laterally to escape it. Under bidirectional conditions ($f = 0$), wakes arrive from opposing directions and no lateral shift helps—regret is inherently limited.

For North Sea planning, this means that sites with strongly unidirectional southwesterly winds (characteristic of the German Bight and southern North Sea) face higher design regret than sites with more bidirectional conditions (such as Danish waters). The DEI wind rose ($a \approx 0.64$, $f \approx 0.38$) falls in the moderate-regret region, consistent with the negligible tradeoffs found by Quick et al. (2026).

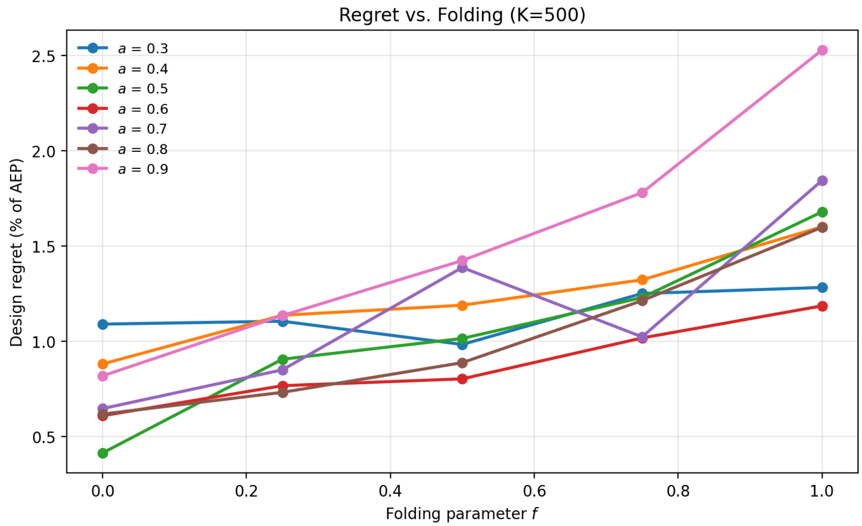

Figure 6. Regret vs. folding f at each concentration a ($K = 500$). Monotonic increase with f is consistent across all a.

3.3 Wind-rose-conditional buffer table

For policy use, we invert the cross-section buffer-decay curves. At each (a, f) on the elliptical-rose grid we ran the cross-section at seven buffer distances $\{2, 5, 10, 15, 20, 30, 40\}D$ with $K_{\text{lib}} = K_{\text{cons}} = 2000$ and the FunWake schedule, took the peak-bearing regret $R(d; a, f)$, and inverted to find the smallest d^* at which $R \leq \tau$. Because the two wake models produce regret magnitudes an order of magnitude apart, each is inverted at thresholds matched to its own scale: $\tau \in \{0.1, 0.2, 0.3\}\%$ AEP for Bastankhah and $\tau \in \{1, 2, 5\}\%$ AEP for TurboPark. Two post-processing choices in this inversion should be stated explicitly because they affect the reported d^*. First, measured $R(d)$ is not monotone in d—multistart noise produces local excursions—so before inverting we take the running maximum from the far end inward, which enforces the monotone decay the policy reading assumes. This is a conservative choice: it can only move d^* outward, and removing it lowers the Bastankhah site values from 7.6–9.3D to 5.8–8.5D. That adjustment is independent of the wake cut-off discussed in Section 2.2, and the two compound: with both removed the Bastankhah site requirement falls to roughly 4–6D. The Bastankhah d^* values reported here should therefore be read as an upper bound, not a point estimate; the TurboPark table, whose entries are censored at $> 40D$ either way, is unaffected. Second, where R never falls below τ within the swept range we treat d^* as *censored* at $> 40D$ rather than extrapolating; censored cells are hatched in the contours and are never interpolated through when reading off site values. Table 2 reports Bastankhah d^* at $\tau = 0.2\%$; Figure 7 shows the corresponding contour with the five real North Sea sites overlaid, and Figure 9 shows the underlying decay curves for both wake models.

3.4 Buffer distance modulates regret

Figure 10 shows regret vs. buffer distance at $K = 500$ for three wind roses.

Table 2. Required buffer distance d^* (in rotor diameters) such that peak design regret $\leq 0.2\%$ AEP, as a function of elliptical-rose parameters (a, f). Bastankhah ($k = 0.04$), $K = 2000$, FunWake schedule; entries marked > 40 denote regret that does not cross the threshold within the swept buffer range. **These values are an upper bound, not a point estimate**: two independent choices—the monotone envelope applied before inversion, and the wake cut-off documented in Section 2.2—each move d^* outward, and removing both lowers the fitted-site requirement from 7.6–$9.3D$ to roughly 4–$6D$. Under TurboPark, even a $10\times$ higher threshold requires buffers beyond $40D$ over most of the shape space (Table 3), and that censoring is insensitive to either choice.

$a \downarrow, f \rightarrow$	0.00	0.25	0.50	0.75	1.00
0.3	4.8	2.8	5.2	4.4	7.1
0.4	2.0	3.5	3.1	2.1	4.5
0.5	2.0	4.3	15.1	4.6	9.9
0.6	2.0	11.3	7.4	15.6	8.3
0.7	11.5	7.2	9.4	18.3	7.0
0.8	2.9	6.5	14.2	23.9	16.3
0.9	3.5	15.1	20.2	17.6	16.7

Table 3. TurboPark buffer table $d^*(a, f; \tau = 2\%)$ in rotor diameters. $K = 2000$, FunWake schedule, TurboPark $A = 0.04$, TI = 0.06. > 40 denotes regret that does not cross the threshold within the swept buffer range.

$a \downarrow, f \rightarrow$	0.00	0.25	0.50	0.75	1.00
0.3	2.0	2.5	10.9	32.9	39.5
0.4	2.0	9.7	15.9	33.9	> 40
0.5	5.5	9.9	34.1	> 40	> 40
0.6	18.8	35.8	> 40	> 40	> 40
0.7	33.0	> 40	> 40	> 40	> 40
0.8	> 40	> 40	> 40	> 40	> 40
0.9	> 40	> 40	> 40	> 40	> 40

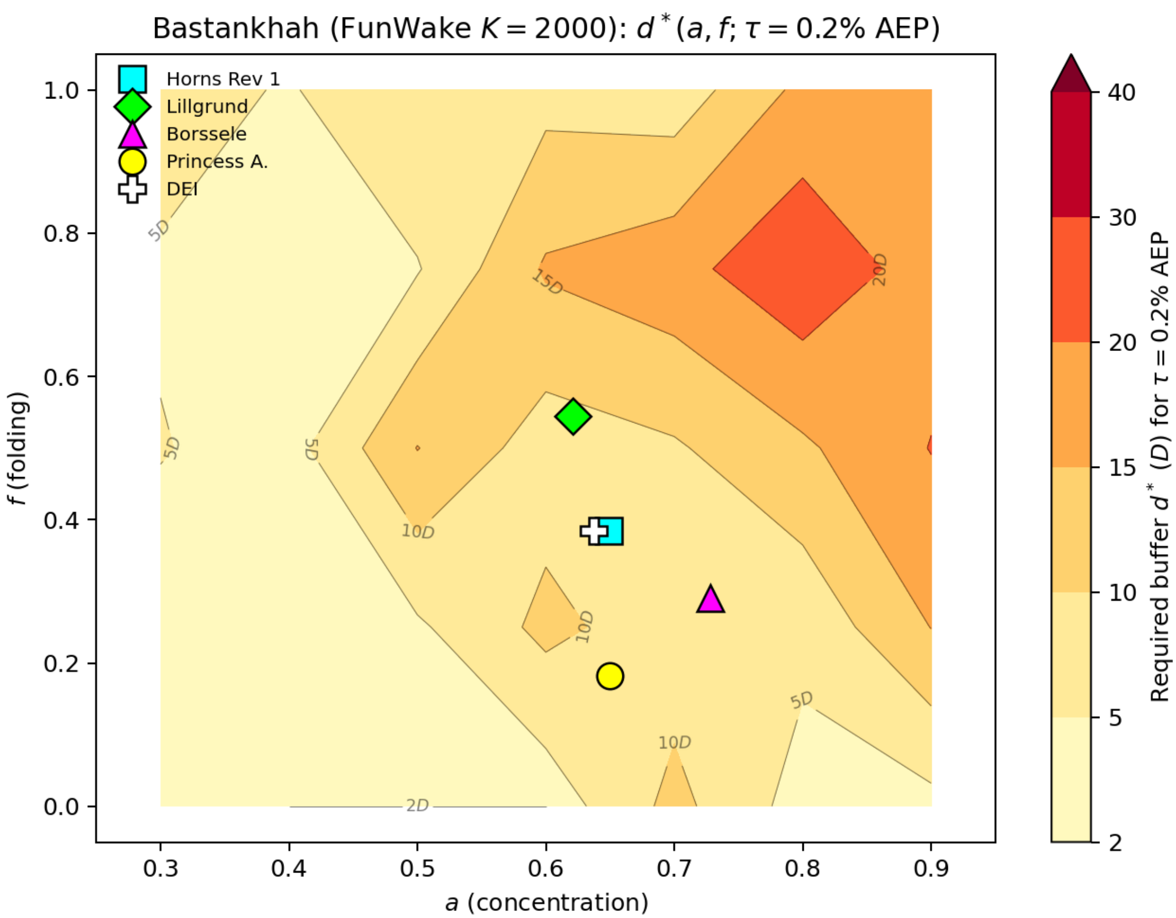

Figure 7. Bastankhah. Required buffer $d^*(a, f)$ at $\tau = 0.2\%$ AEP ($K = 2000$, FunWake schedule). Markers: fitted real-site placements (Horns Rev 1, Lillgrund, Borssele, Princess Amalia, DEI). The five fitted real-site placements require $d^* \approx 8$–$9D$ ($\sim$2 km) at this threshold.

For concentrated unidirectional wind ($a = 0.9,\ f = 1.0$), regret drops from 2.5 % at $2D$ to 0.4 % at $40D$ (9.6 km). The bidirectional case ($a = 0.5,\ f = 0.0$) shows gentle decay from 0.4 % to 0.2 %.

Under the Bastankhah model, the NYSERDA 4 nmi ($\sim 31D$) recommendation reduces regret to <1 % for all wind roses tested, but is unnecessarily conservative for bidirectional sites. One caveat on that comparison: the NYSERDA figure is a nearest-turbine separation, whereas our buffers are boundary-gap distances between lease polygons, so the two are not measured between the same points and the comparison is indicative rather than exact. Under TurboPark the same recommendation is insufficient rather than conservative: the converged buffer table (Table 3) places the $\tau = 2\%$ requirement beyond $40D$ for all five fitted real sites. Wind-rose-dependent buffer standards could allow denser development at low-risk sites (bidirectional, e.g., DEI) while maintaining protection at high-risk sites (unidirectional, e.g., German Bight)—but the adequacy of any fixed

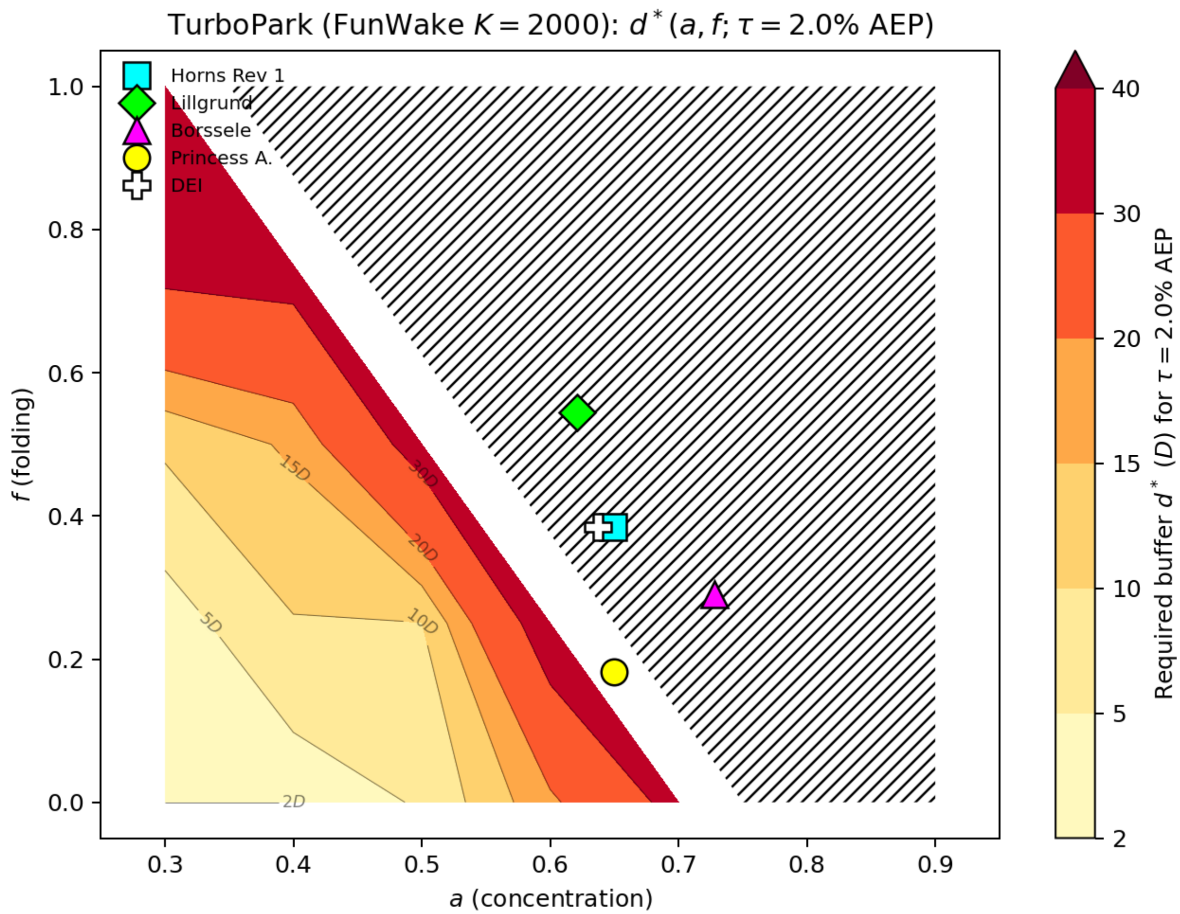

Figure 8. TurboPark. Required buffer $d^*(a, f)$ at $\tau = 2.0\%$ AEP under TurboPark ($A = 0.04$, TI = 0.06; $K = 2000$, FunWake schedule). Note the threshold is $10\times$ higher than Fig. 7, because TurboPark's slower wake recovery raises realistic-scenario regret by one to two orders of magnitude. Hatching marks cells where the threshold is never reached within the swept range, so d^* is censored at $> 40D$ rather than measured. That censored region covers most of the shape space, including every one of the five fitted real sites; we therefore report their requirement as $> 40D$ (> 9.6 km) and deliberately do not interpolate a finite value through censored cells.

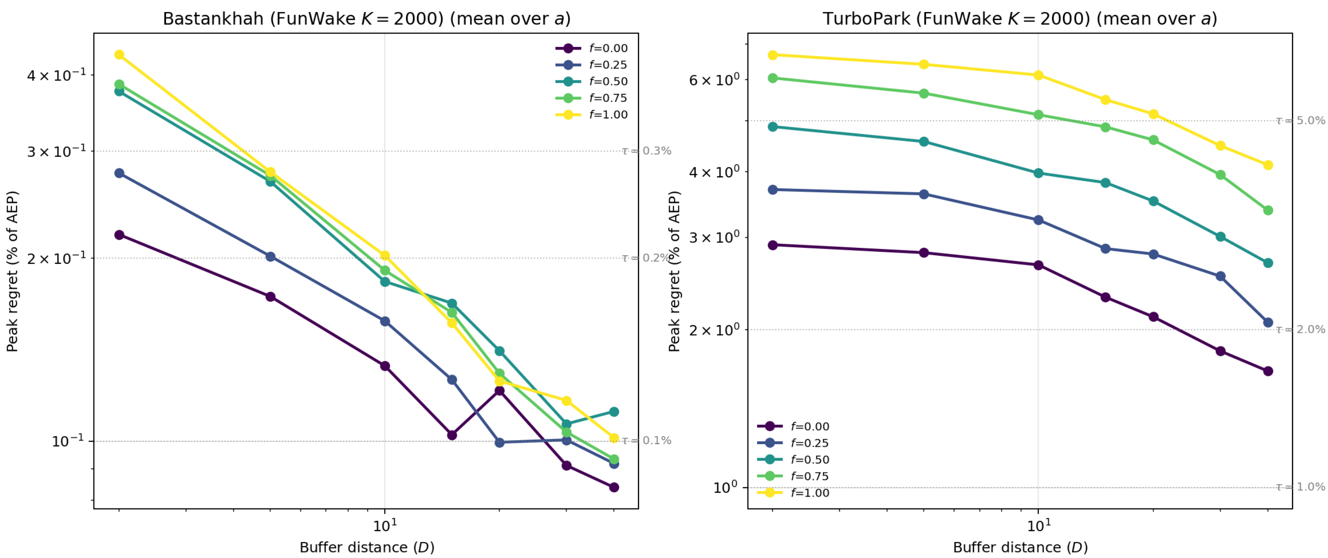

Figure 9. Buffer-decay curves $R(d)$ averaged over a at each folding value f, for Bastankhah (left) and TurboPark (right); $K = 2000$, FunWake schedule. Reference τ thresholds (dotted) are matched to each model's regret scale (0.1–0.3% for Bastankhah, 1–5% for TurboPark).

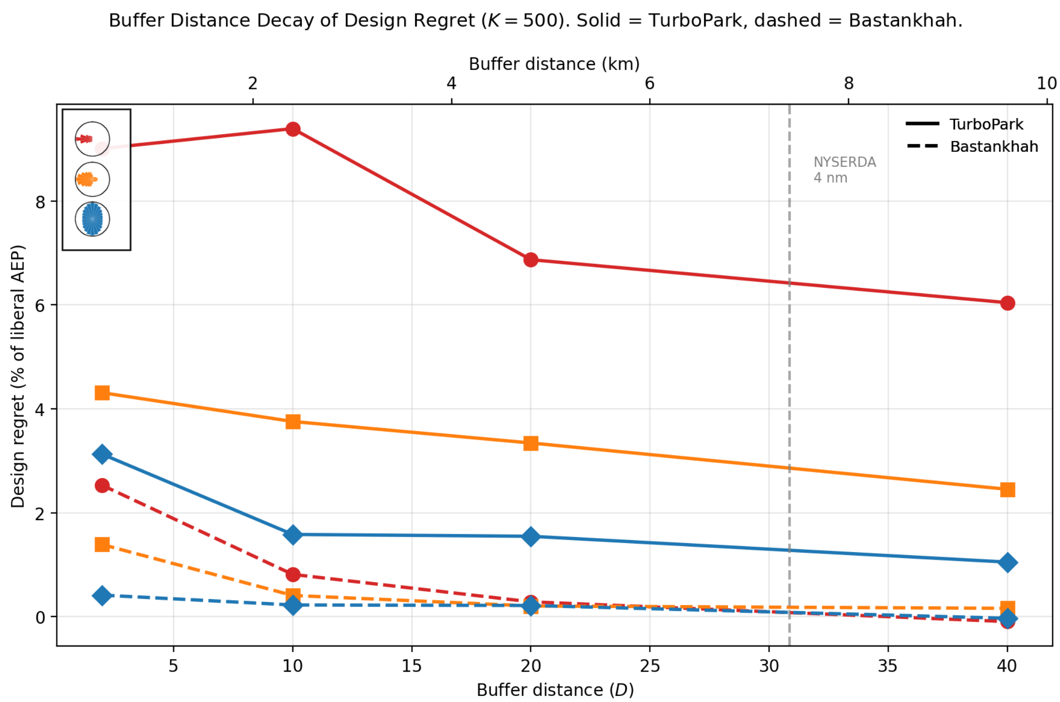

Figure 10. Regret vs. buffer distance at $K = 500$, expressed as % of liberal AEP. The NYSERDA 4 nmi recommendation is marked.

standard is wake-model-dependent. This is directly relevant to the emerging cross-border coordination frameworks for the North Sea (Fliegner et al., 2025).

3.5 Directional vulnerability: regret cross-sections

The greedy grid search (Sections 3.2 and 3.4) establishes an adversarial upper bound by placing individual unconstrained turbines. To assess regret under more realistic conditions, we now turn to the regret cross-sections, in which an identical copy of the target farm is placed at controlled boundary-gap distances around the target (Section 2.5).

3.5.1 Realistic multi-directional cases.

Figures 11 and 12 present cross-sections for six multi-directional wind roses under both wake models: the DEI real wind rose and five elliptical parameterizations (Table 4). Under Bastankhah, peak regret from a realistic identical neighbor farm is modest: 0.16–0.49 % of AEP, with all peaks at the closest buffer distance ($2D$). Under TurboPark the same scenario produces 1.9–10.6 % of AEP—an order of magnitude larger, and for the concentrated unidirectional rose ($a = 0.9$, $f = 1.0$; 10.6 %) *exceeding* the greedy adversarial bound computed with the baseline optimizer (9.0 %, Section 3.2). This is a direct demonstration that "upper bounds" from adversarial placement are bounds only relative to the optimizer configuration used to compute them: a stronger inner optimizer applied to a merely realistic scenario can overtake a weaker optimizer's adversarial ceiling. Under Bastankhah the greedy bound (0.4–2.5 %) does exceed the realistic peaks (0.16–0.49 %) by roughly $5\times$. Regret decays with buffer distance for all cases; Figures 13 and 14 recast the same data as peak regret over the (a, f) shape space at fixed buffer distances. The directional pattern also differs between models: under TurboPark, peak regret sits at or within 15° of the prevailing upwind direction for four of the six cases; under Bastankhah, several peaks sit 45–60° oblique to it (e.g. 330° for DEI and for $a = 0.9$, $f = 1.0$), consistent with the liberal layout being partially pre-adapted to the dominant direction and most vulnerable on its flanks. The single-direction stress test below makes this contrast unambiguous.

Real clusters rarely stop at one neighbor; Section 3.6 examines how regret scales with the *number* of neighboring farms.

3.5.2 Validation and stress test: single-direction case.

As a methodology check we also ran a degenerate single-direction case (all energy from 90°). Regret is precisely zero for all downwind bearings, as it must be when the neighbor casts no wake over the target. Under Bastankhah, peak regret reaches 1.26 % of AEP at the closest buffer, at a bearing oblique to the inflow (135°)—the same flank-vulnerability pattern as the multi-directional cases. Under TurboPark the peak is 38 % of AEP and sits directly upwind (90°): with slowly-recovering wakes, no lateral rearrangement rescues the layout from a neighbor parked in the single energy corridor, so the oblique-peak ("ambush") structure is a Bastankhah-specific consequence of fast wake recovery rather than a universal feature. The qualitative pattern and the flow-field interpretation are reported in Appendix A, with a representative flow map (135°/$40D$) showing how a southeastern neighbor's wakes drift westward and graze the target's southern row.

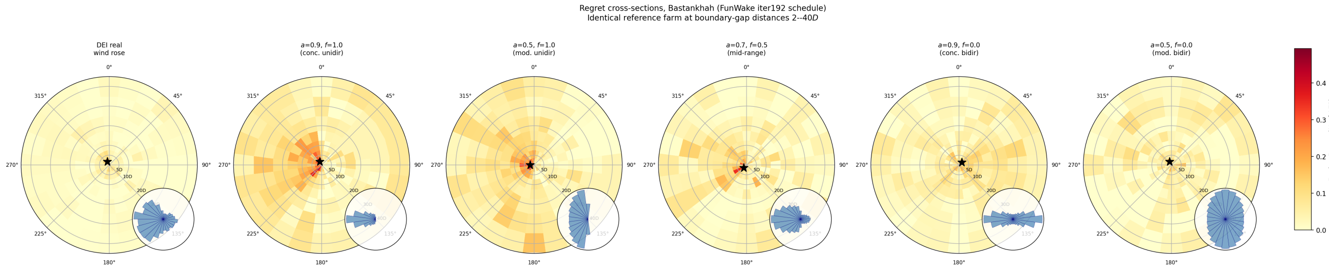

Figure 11. Bastankhah. Regret cross-sections for six multi-directional wind roses ($K = 2000$ for the elliptical roses, $K = 500$ for DEI beyond $15D$; FunWake schedule; boundary-gap distances). Each panel shows design regret (% of AEP) as a function of neighbor bearing and buffer distance for an identical copy of the target farm. Stars mark peak regret. Inset wind roses show the sector frequency distribution. All panels share the same color scale.

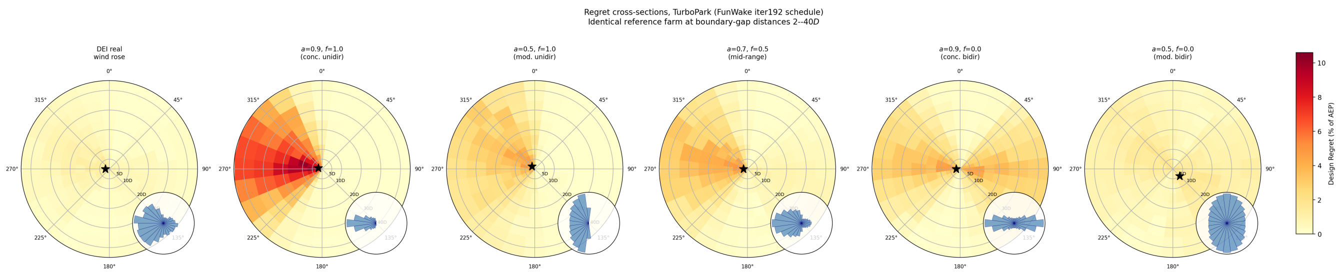

Figure 12. TurboPark. Same cross-sections as Fig. 11 under the TurboPark wake model. Peak regret is an order of magnitude larger (note the color scale) and sits at or near the prevailing upwind direction in most cases, in contrast to the oblique peaks under Bastankhah.

Table 4. Peak regret from cross-sections under both wake models (FunWake schedule; boundary-gap distance). Elliptical roses at $K = 2000$; DEI mixes $K = 2000$ (near distances) and $K = 500$ (far distances); single-direction case at $K = 500$.

	Bastankhah				TurboPark			
Case	GWh	%	Bearing	Buffer	GWh	%	Bearing	Buffer
DEI real wind rose	8.9	0.16	330°	$2D$	108	1.9	270°	$2D$
$a = 0.9$, $f = 1.0$	19.8	0.49	330°	$2D$	426	10.6	285°	$2D$
$a = 0.5$, $f = 1.0$	15.3	0.38	270°	$2D$	194	4.9	315°	$2D$
$a = 0.7$, $f = 0.5$	15.1	0.38	225°	$2D$	218	5.4	270°	$2D$
$a = 0.9$, $f = 0.0$	9.8	0.24	45°	$2D$	193	4.8	270°	$2D$
$a = 0.5$, $f = 0.0$	7.4	0.19	330°	$2D$	80	2.0	135°	$5D$
Single direction (90°)	51.0	1.26	135°	$2D$	1549	38.2	90°	$2D$

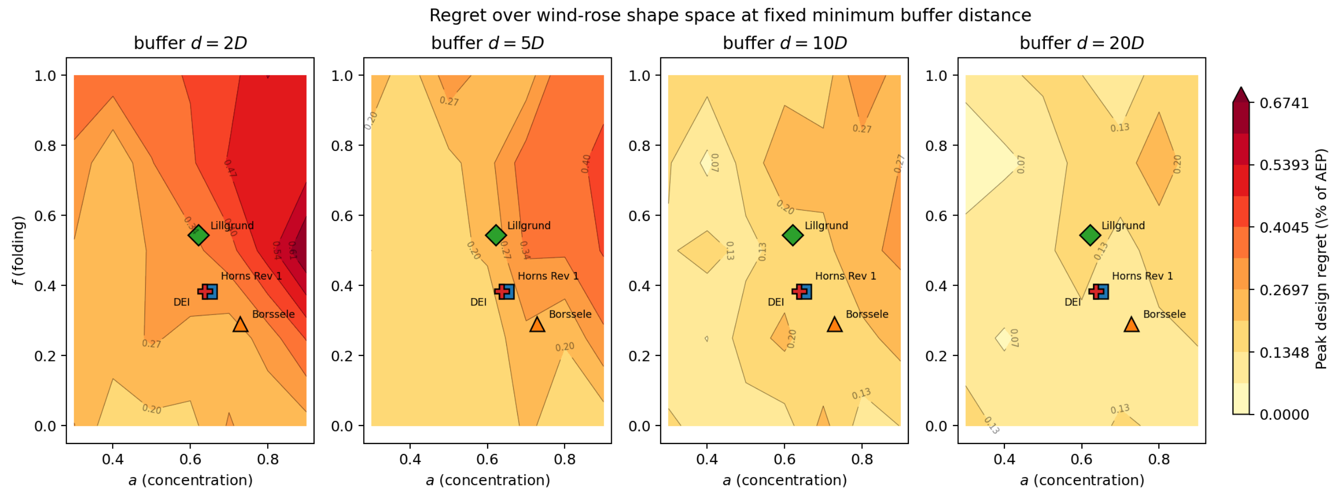

Figure 13. Bastankhah. Alternative view of the cross-section data (*constrained, identical-copy neighbor farm*—not the unconstrained greedy upper bound): peak regret over the elliptical-rose shape space (a, f) at four fixed minimum buffer distances ($K = 2000$, FunWake schedule). Real-site fits overlaid as labelled markers. Hot corner (high concentration a + high folding f, i.e. concentrated unidirectional) shrinks with buffer; operational North Sea sites sit in the cool moderate-shape region for all buffer choices.

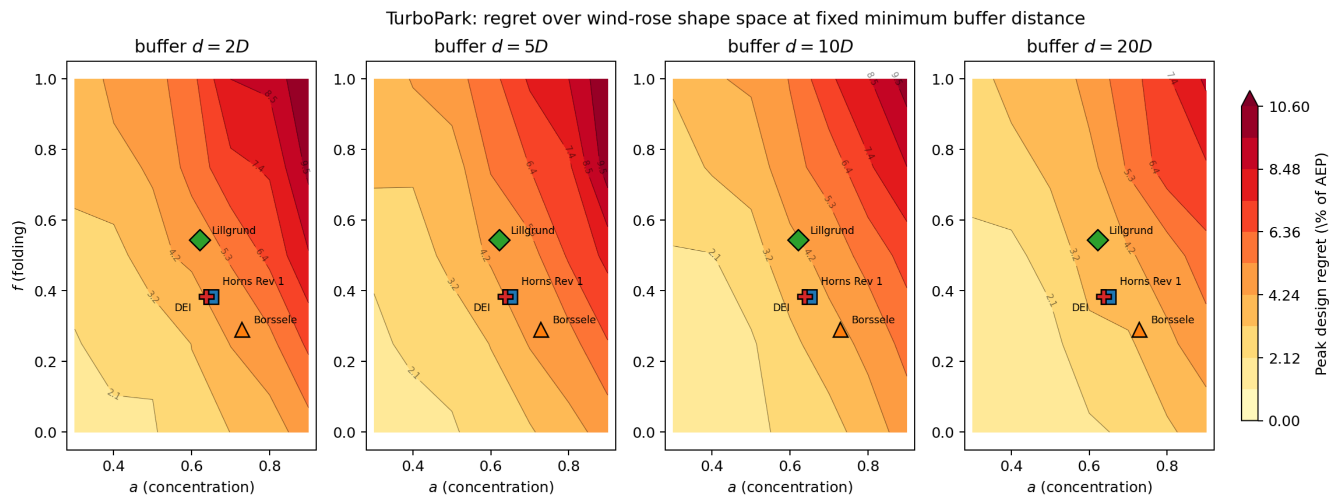

Figure 14. TurboPark. Same view as Fig. 13 but under the TurboPark wake model ($K = 2000$, FunWake schedule). Peak regret runs an order of magnitude higher across the shape space, the hot corner persists at all buffers shown, and the real sites no longer sit in a uniformly cool region. The wake-model bracket on realistic regret dominates any single number.

3.5.3 a-ablation and the geometry of the elliptical rose.

To trace how regret varies with rose concentration, Figure 15 sweeps a from 0.01 to 0.9 at fixed full folding ($f = 1$), with the DEI real rose overlaid for reference (baseline schedule, $K = 500$; values in this paragraph are from that configuration and sit somewhat below the converged FunWake values in Table 4). The parameterization has a non-obvious structure: $a = 1/\sqrt{\pi} \approx 0.564$ gives a uniform rose; $a > 0.564$ elongates the ellipse along the prevailing direction, concentrating energy there (at $a = 0.9$, $f = 1$, 20.6 % of energy arrives from 270°); $a < 0.564$ elongates it perpendicular to the prevailing direction, producing a bimodal rose with peaks at $\theta_{\text{prev}} \pm 90°$ (at $a = 0.01$, $f = 1$, ∼50 % each at 0° and 180°). Small a therefore does not approximate the single-sector stress test; it represents a distinct perpendicular-bimodal regime. Regret reflects this: tiny a (0.01–0.05) reaches 1.5–1.8 % at $2D$ because the target faces two concentrated inflow axes; moderate a (0.3–0.5, near-uniform) sits at 0.1–0.3 %; and $a = 0.9$ (concentrated westerly) yields ∼0.3 %. Realistic North Sea wind climates sit in the moderate-a valley. The perpendicular-bimodal and single-sector corners of the rose space both produce ≥1 % regret, but they represent physically distinct wind climates; neither is characteristic of operational offshore sites.

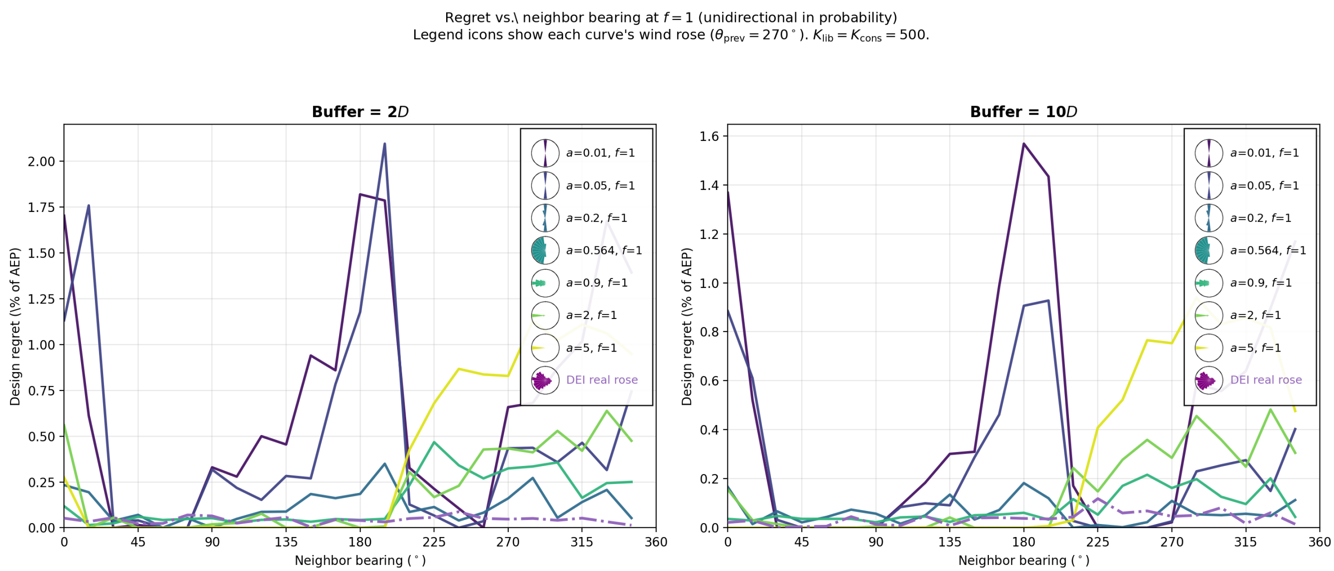

Figure 15. Regret vs. neighbor bearing as the rose concentration a varies at full folding ($f = 1$, prevailing direction 270°; baseline schedule, $K = 500$). Colored curves: a-sweep from perpendicular-bimodal ($a \ll 0.564$) through uniform ($a \approx 0.564$) to concentrated unidirectional ($a = 0.9$), with miniature wind-rose icons in the legend. The DEI real wind rose is overlaid for reference. Left: $2D$ buffer; right: $10D$.

3.6 Regret with multiple neighboring farms

Most operating clusters expose a farm to several neighbors at once. To isolate how regret scales with the number of neighbors n—without the path-dependence of a greedy placement sequence—we place n identical copies of the liberal layout at evenly spaced bearings starting from the prevailing upwind direction, each at the same nominal boundary gap, for $n = 1$–8 (FunWake schedule, $K = 500$, three wind roses, both wake models, nominal gaps $2D$ and $10D$). Copies must also respect a $2D$ boundary gap to *each other*; when a ring cannot satisfy this, all offsets are scaled up by the smallest common factor that does (found by bisection), and the realized gaps are recorded. This inter-neighbor minimum is a modelling choice rather than a physical constant, so Appendix F sweeps it over a forty-fold range and confirms that no conclusion below depends on it. Figure 16 shows regret, AEP loss, and their ratio (the recoverable fraction of neighbor-induced damage) versus n.

It is useful to read these results through the identity

$$R = \underbrace{\left[\mathrm{AEP}(\mathbf{x}^*_{\mathrm{lib}}) - \mathrm{AEP}(\mathbf{x}^*_{\mathrm{lib}}, \mathbf{n})\right]}_{\text{AEP loss } L} \times \underbrace{R/L}_{\text{recoverable fraction}}, \tag{6}$$

which separates how much damage the neighbors do from how much of it a redesigned layout can avoid. Design regret is not a measure of damage; it is the value of foresight, and foresight is worth nothing against damage that cannot be dodged.

Three findings emerge. First, while the farms can physically pack at the nominal gap ($n \le 4$ for DEI-sized farms), regret *grows* with n but strongly sublinearly: under TurboPark at $2D$, four ringed farms produce 1.2$\times$ (concentrated unidirectional), 1.7$\times$ (DEI), and 3.0$\times$ (moderate bidirectional) the single-farm regret—never $4\times$. The growth factor is largest for bidirectional roses, whose multiple energy-carrying directions each admit a damaging neighbor; under multi-neighbor conditions the bidirectional advantage seen in the single-neighbor analyses partially erodes. Second, the two factors in Eq. (6) move in opposite directions as neighbors accumulate: loss rises to $n = 4$ while the recoverable fraction falls throughout (TurboPark, concentrated unidirectional: ≈0.6 for a single upwind farm to 0.38 at $n = 8$). This is why regret grows so much more slowly than the damage does, and it is a property of the decomposition rather than of the sampling. Third, the sharp *decline* beyond $n = 4$ is a separate, geometric effect and should not be read as a continuation of the same mechanism: five or more DEI-sized farms cannot all sit close to the target, so the feasible ring jumps outward (realized gaps of $40D$ and beyond) and regret falls steeply with distance (12.2 % at $n = 4$ to 3.5 % at $n = 8$). The turnover in n is therefore real for a planner—who is likewise bound by lease geometry—but it is a consequence of what can be built rather than of the wake physics alone. Section 3.8 shows this packing limit follows from farm size and is robust to our spacing rule. A greedy adversarial variant of this experiment (Appendix D) corroborates the sublinear growth with a weaker optimizer configuration.

3.6.1 What closes the escape routes.

The natural reading of the falling recoverable fraction is that *symmetry* is responsible: a target attacked from every side has nowhere to move. That reading is wrong, and a controlled experiment separates the two candidate explanations. Holding the total neighboring capacity, the total neighboring area and the buffer gap all fixed, we divide that capacity into n smaller farms spread around the target (each the target polygon scaled by $1/\sqrt{n}$ carrying $\sim N_t/n$ turbines on a regular grid: 50, 25, 17, 12 and 10 turbines for $n = 1$ to 5), which keeps every ring at the nominal $2D$ gap throughout. This spreads the threat around the compass while adding nothing to it. The recoverable fraction *rises* with n rather than falling: under TurboPark monotonically from 0.51 at $n = 1$ to 0.65 at $n = 5$ (concentrated unidirectional) and from 0.19 to 0.30 (moderate bidirectional), the opposite of the trend in the identical-copy ring. Meanwhile the AEP loss follows the wind: spreading raises it under a bidirectional rose

(3.9 % at $n = 1$ to 7.0 % at $n = 5$) and lowers it under a unidirectional one (12.6 % to 5.6 % at $n = 3$, recovering to 7.2 % at $n = 5$), because wake is being moved into or out of the directions that carry energy. Under Bastankhah the same experiment points the same way for the unidirectional rose (0.21 to 0.32) but is not resolved for the bidirectional one, where the pooling clip binds at $n = 2$—the noise-floor behaviour documented in Section 3.1. Because these neighbors are grid layouts rather than liberal-optimized copies, their absolute magnitudes are not directly comparable with the identical-copy ring—an optimized neighbor is the more damaging one, costing 17.7 % against 12.6 % at $n = 1$—so only the within-experiment trend should be read across the two constructions.

The governing quantity is therefore not angular symmetry but the *concentration of wake mass in the directions that carry energy*. Adding full-size farms piles mass into those directions and closes escape routes; dividing a fixed capacity narrows every individual wake corridor and opens them; concentrating the wind onto a single axis leaves one corridor that a redesigned layout can slide out of. All three interventions, and the single-neighbor wind-rose result of Section 3.2, follow from this one statement.

For policy, worst-case single-neighbor regret multiplied by a small factor ($\approx$3 at the bidirectional extreme, $\approx$1.2 for concentrated unidirectional roses) bounds the dense-cluster case, and the bound tightens further once realistic lease geometry precludes close packing of many neighbors.

3.7 How much of a regret number is the optimizer?

Every regret value in this paper is the output of two nested optimizations, so it is worth measuring directly how much of it the inner solver determines. We do this by holding a neighbour configuration completely fixed—the 30 turbines placed by a completed greedy search—and re-computing regret twice, once with each schedule, on the same multistart set. Only the inner optimizer differs.

Across four wind roses and both wake models, the tuned schedule raises measured regret in every case, by 1.10–2.13$\times$ (median 1.41$\times$); the largest effects appear where regret is smallest. Under-optimization therefore depresses these estimates, which is the direction that makes an inadequately optimized study look reassuring.

Two cautions follow, and they cut against a naive reading of that ratio. First, the sign is not guaranteed. A weak *conservative* search lowers regret, but a weak *liberal* search finds a layout that is more exposed to neighbours and so raises it; which dominates is configuration-specific, and we observe both (the multi-farm experiment of Appendix D, run at a lower budget, reports *higher* regret than its converged counterpart). “We under-optimized, therefore we are conservative” is not a valid inference in either direction. Second, this experiment measures the optimizer’s effect at a fixed configuration; it does not establish what a stronger search would have *found*. A greedy configuration is co-adapted to the liberal layout it was built against, and because that layout is not stored it cannot be reused: re-evaluating the same neighbours against an equally good but different liberal optimum reproduces the reported value within 25 % in six of eight cases but not in the other two. Adversarial-search outputs should accordingly be read as “the worst found under this configuration with this optimizer”, not as bounds.

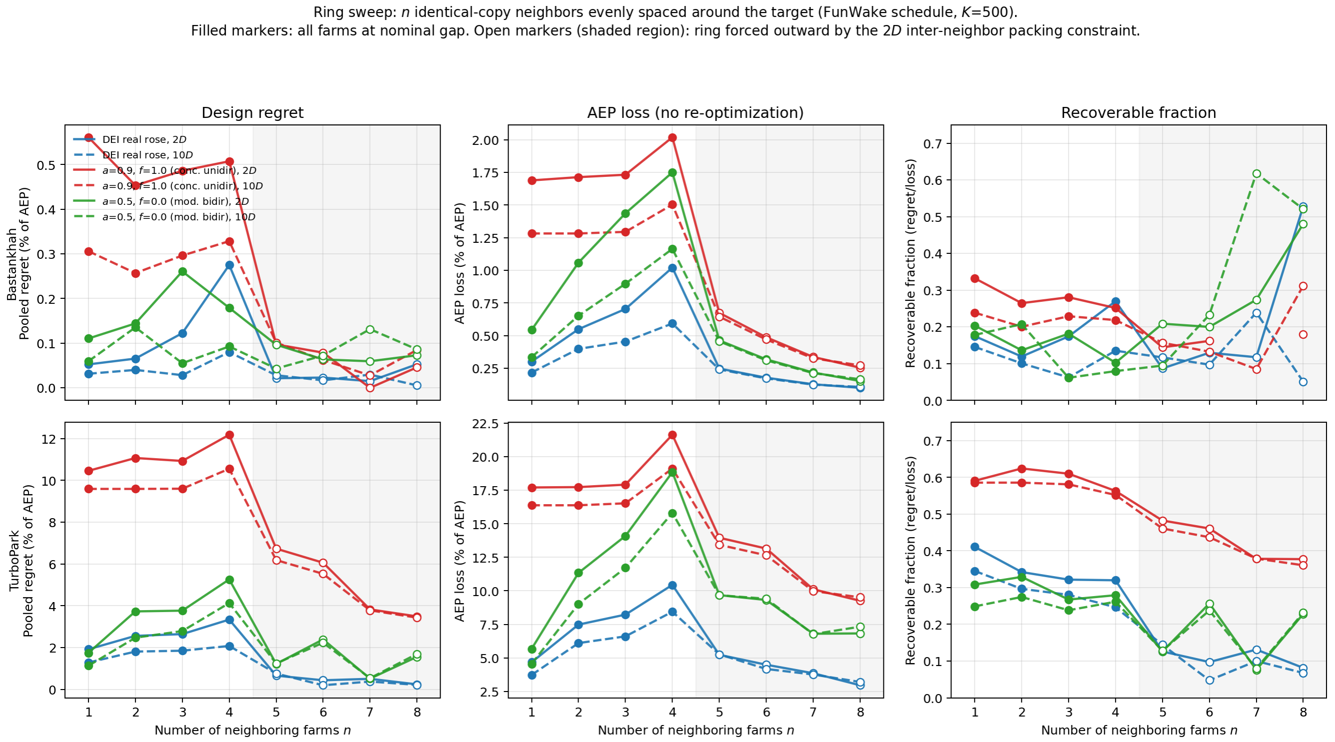

Figure 16. Ring sweep: n identical-copy neighbors evenly spaced around the target, starting from the prevailing upwind bearing (FunWake schedule, $K = 500$). Columns: pooled regret, AEP loss of the liberal layout, and their ratio (recoverable fraction). Rows: Bastankhah (top), TurboPark (bottom). Filled markers: all farms at the nominal boundary gap. Open markers (shaded region): the $2D$ inter-neighbor packing constraint forces the ring outward, so realized gaps exceed the nominal value.

3.8 Lease geometry bounds the multi-neighbor problem

The packing limit encountered above is not an artifact of our spacing rule but a property of farm size, and it can be established without any wake computation. A ring of n neighbors must clear both the target and each other. Because a 50-turbine farm in the DEI polygon is itself $\sim 150D$ across (mean radius $71D$), the ring radius is dominated by the farms' own footprint rather than by the buffer: at a $2D$ buffer the buffer contributes only $\sim 3\,\%$ of it. Figure 19 sweeps the requested buffer and reports the largest ring for which every smaller ring also fits. At the $2D$ inter-neighbor minimum used throughout, four is the largest ring that fits anywhere in the 2–$40D$ range this study sweeps; a fifth first fits at $42D$ (10 km), a sixth at $68D$ (16 km), and a seventh at $98D$ (24 km). A simple model, $n \approx \pi/\arctan(R_{\text{farm}}/R_{\text{ring}})$ with $R_{\text{ring}} = R_{\text{farm}} + d$, captures the scaling and is exact at $n = 4$, though it overestimates capacity by 0.6–0.9 farms at the larger rings because it treats the footprint as a disc.

The consequence for cluster planning is that the alarming corner of the parameter space—many neighbors, all close—is not merely unstudied, it is geometrically unbuildable with same-sized farms. By the time six neighbors fit, they sit 16 km away in boundary gap—35–53 km centroid to centroid—where regret has already collapsed. Dense encirclement at close range is

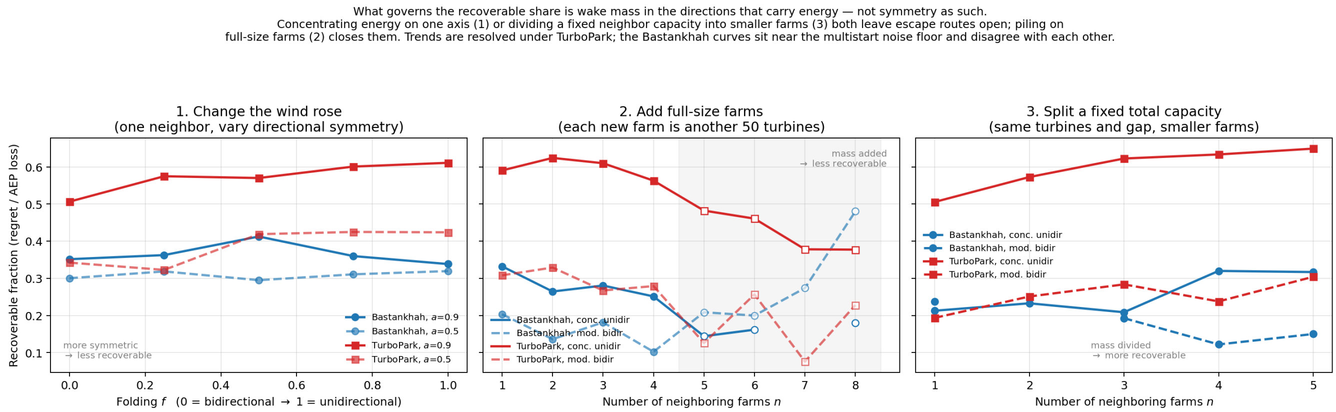

Figure 17. Three interventions on the same quantity, all under TurboPark (Bastankhah shown for completeness; its curves sit near the multistart noise floor). (1) Concentrating the wind onto one axis, (2) adding full-size neighboring farms, (3) dividing a fixed total neighbor capacity into more, smaller farms at fixed buffer gap. Symmetry alone does not predict the recoverable fraction: intervention (3) spreads the threat around the compass yet raises it. Concentration of wake mass in the energy-carrying directions does.

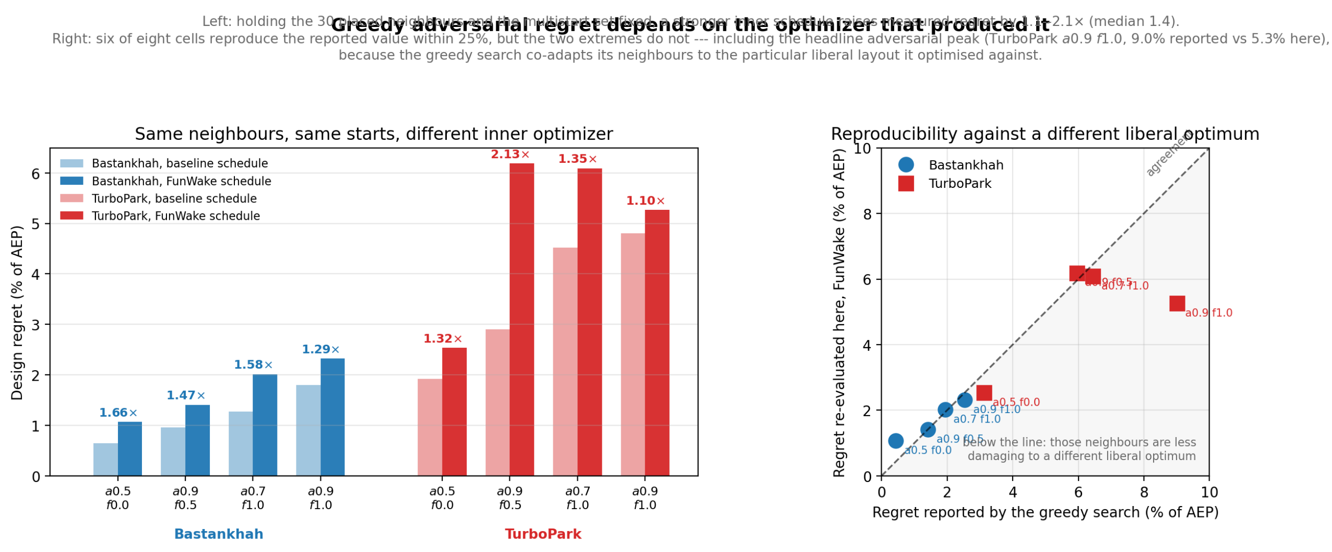

Figure 18. Optimizer effect at fixed neighbour configuration. Left: the same 30 placed neighbours and the same multistart set, evaluated under both schedules; the tuned schedule raises measured regret by 1.1–2.1×. Right: the same re-evaluation against the originally reported value, showing that most—but not all—cells reproduce.

realistic only when the neighbors are substantially smaller than the target, which is precisely the divided-capacity configuration examined above.

Two caveats attach to this construction. The steps in Figure 19 are not monotone in the buffer: for an elongated footprint, equal *boundary* gaps place farms at very unequal centroid radii ($85D$ to $177D$ at a $20D$ buffer), so neighbor-pair clearances rise and fall as the ring grows, and a given n can fit at one buffer, fail at a larger one, and fit again beyond that. For the same reason our rings are evenly spaced in bearing but not in centroid distance. Both are consequences of using a realistic irregular concession polygon rather than a disc.

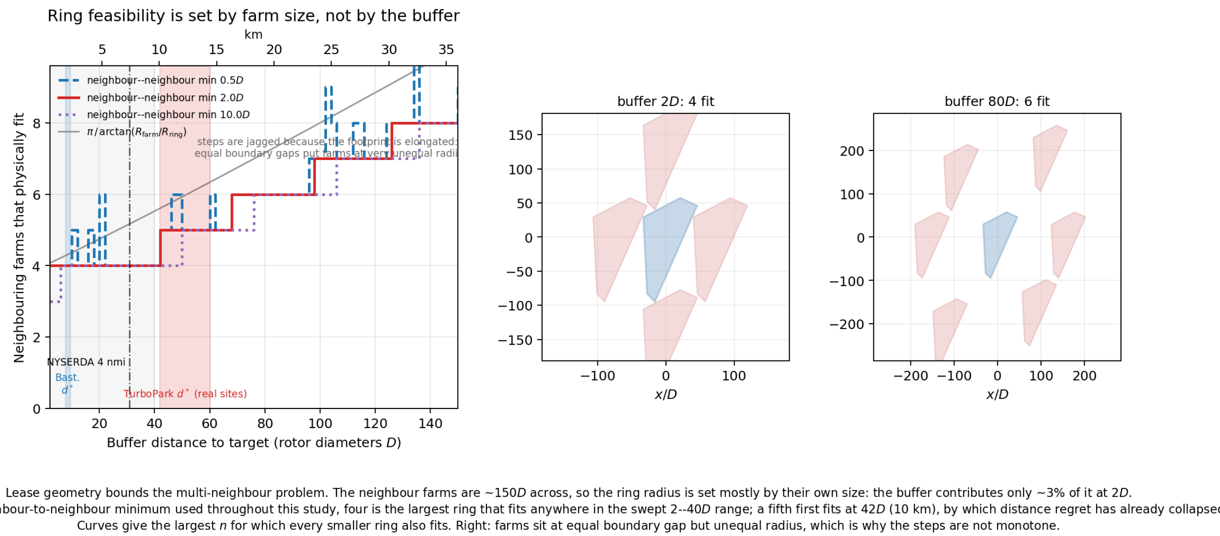

Figure 19. Geometric feasibility of multi-neighbor rings—no wake model involved. Left: largest ring size for which every smaller ring also fits, versus requested buffer, for several inter-neighbor minimum gaps; grey curve is the analytic estimate, grey band the buffers swept in this study. Right: the arrangements at $2D$ and $80D$, showing that equal boundary gaps place farms at unequal radii.

4 Discussion

4.1 Real North Sea sites in the shape space

To ground the parametric results in real planning decisions, we fit the elliptical model to five North Sea offshore wind sites (Table 5, Fig. 5). All five sites fall in the moderate-regret region of the (a, f) space, with $a \in [0.62, 0.73]$ and $f \in [0.18, 0.54]$. No operational North Sea site in our sample reaches the high-regret corner ($a > 0.8$, $f > 0.75$), though this does not preclude future lease areas—particularly in the southern North Sea—from doing so.

The sites cluster in the (a, f) space because North Sea wind climates share a common driver: Atlantic weather systems producing dominant southwesterly winds. The spread in f (0.18–0.54) reflects varying degrees of bidirectionality, with Lillgrund the most unidirectional ($f = 0.54$, highest regret risk among the five). The Princess Amalia rose is poorly captured by the

Table 5. Fitted elliptical wind rose parameters for North Sea offshore sites. Data sources: Horns Rev 1 and Lillgrund from PyWake; Borssele from IEA Wind Task 55 ROWP; Princess Amalia from the python-windrose package; DEI from 10-year NEWA timeseries.

Site	a	f	θ_{prev}	R^2
Horns Rev 1 (DK)	0.649	0.384	240 °	0.82
Lillgrund (SE)	0.621	0.544	236 °	0.77
Borssele (NL)	0.728	0.291	240 °	0.87
Princess Amalia (NL)	0.650	0.182	210 °	0.49
Danish Energy Island	0.637	0.384	251 °	0.84

elliptical model ($R^2 = 0.49$), reflecting a residual third mode that the two-parameter ellipse cannot represent; the fitted values should be read as an approximate placement in the shape space rather than a faithful reproduction of the rose. A mixture-of-ellipses extension (Hart, 2025) would be needed for full fidelity at sites with three or more directional modes; Appendix B explores regret under such mixture roses and finds it stays within the single-rose envelope.

4.2 Implications for North Sea cluster development

The six-fold variation in design regret across wind rose shape has direct implications for the North Sea's emerging cluster framework.

4.2.1 Economic magnitude.

For a representative 750 MW farm (50×15 MW), the realistic-scenario regret at a typical North Sea rose (DEI) spans the wake-model bracket 0.16 % (Bastankhah) to 1.9 % (TurboPark) of AEP at close spacing, i.e. ∼9–108 GWh/yr against our modelled AEP. That denominator is itself optimistic—our speed treatment implies an 84.7% capacity factor for DEI (Section 4.4)—so on a realistic 50–55% capacity factor the same percentages correspond to roughly 6–70 GWh/yr, and the figures below should be scaled accordingly. At a contract-for-difference strike price of 70 EUR/MWh, this is ∼0.6–7.6 M EUR/yr, or—undiscounted, and therefore an upper figure—∼16–190 M EUR over a 25-year operational lifetime; at a 6 % discount rate the present value is roughly half that. Even the low end of this bracket is order-of-magnitude comparable to the cost of running a neighbor-aware layout optimization study, which is negligible relative to project CAPEX; the high end, and the concentrated-unidirectional worst case (10.6 % under TurboPark), represent project-material losses. The headline economic case is therefore: the cost of doing the analysis is small enough that even the most optimistic wake model justifies it, and under slower-recovery wake physics the stakes are tens to hundreds of millions of euros per farm.

4.2.2 German Bight and southern North Sea.

Sites with strongly unidirectional southwesterly winds sit in the high-regret region ($f \gtrsim 0.75$). Here, the value of coordinated design is highest: buffer distances of 30–$40D$ (7–10 km) are needed to bring the greedy adversarial bound below 1 % under

Bastankhah, and under TurboPark even $40D$ leaves several percent of AEP at stake in the realistic identical-copy scenario (Table 3). The projected 30 % capacity factor reduction from cluster wakes (Fliegner et al., 2025) is partially a design problem, not just a physics problem: some of this loss could be mitigated by neighbor-aware layout optimization.

4.2.3 Danish waters.

The DEI wind rose ($f \approx 0.38$) benefits from a natural resilience: wakes from any single direction are partially offset by energy captured from the opposing direction, limiting the rearrangement advantage of a neighbor-aware design. This explains the negligible design tradeoffs found by Quick et al. (2026). However, this resilience is a property of the wind climate, not the farm—other Danish sites with different wind characteristics may not share it.

4.2.4 Cross-border coordination.

The value of inter-farm coordination is itself wind-rose-dependent. Our results suggest that coordination frameworks like the "good neighbour" principle in the UK EN-3 guidance should be weighted by wind rose directionality. Fixed buffer distances—such as the NYSERDA 4 nmi recommendation—are simultaneously inefficient for bidirectional sites (which need only $\sim 10D$ under Bastankhah) and insufficient for all sites under TurboPark (Table 3). Wind-rose-conditional buffers could allow denser development at low-risk sites without compromising high-risk ones, but setting the absolute scale requires committing to a wake model.

4.3 The convergence problem

Our most broadly applicable finding is methodological. Small multistart budgets—even $K = 100$—can miss substantial regret, and the required budget depends on the size of the regret signal relative to multistart noise: large signals (TurboPark) are captured to within a few percent by $K = 500$, while near-noise-floor signals (Bastankhah realistic scenarios) are not converged even at $K = 2000$ (Fig. 4). The cause is catastrophic cancellation: regret is a difference between two large, noisy AEP values. A 0.5 % miss in either optimization can produce a 50 % error in the difference. The convergence budget is moreover conditional on the inner optimizer: a stronger iteration schedule raises the regret estimate at every K, so "converged in K" does not imply "converged" in any absolute sense.

This affects any study computing AEP differences between strategies. We recommend $K \geq 500$ for 50-turbine farms, storing per-start AEPs so that convergence can be verified by subsampling after the fact, and reporting the capture fraction alongside the regret estimate. The greedy sweep at $K = 500$ took a median of 27 hours per wind rose on a single AMD MI250X GCD, or roughly 950 GPU-hours for the full 35-case sweep; the $K = 2000$ FunWake production grid cost roughly 22 (Bastankhah) to 37 (TurboPark) GPU-hours per (a, f, d) cell.

4.4 Limitations

4.4.1 Wake model.

Our primary sweep uses Bastankhah and Porté-Agel (2014) with $k = 0.04$ and root-sum-of-squares superposition, and every cross-section and buffer-table analysis is replicated in full under TurboPark (Nygaard et al., 2022) with TI = 0.06. The Bastankhah realistic-scenario magnitudes (0.16–0.49 % AEP) sit at or below the absolute uncertainty of typical engineering wake models, so they should be read as bounds on *the optimization-recoverable component of inter-farm losses*, not as predictions of measured AEP differences. Under TurboPark the same scenarios produce 1.9–10.6 % of AEP (Table 4, Appendix G): the two models bracket realistic-scenario regret over an order-of-magnitude range. The qualitative findings—folding dominance, monotone buffer decay, the multistart convergence requirement—are shared by both models, but the directional vulnerability pattern is not: the oblique-peak structure appears only under Bastankhah's fast wake recovery (Section 3.5). Neither model is closer to ground truth—they are both engineering parameterisations with different recovery assumptions—so the realistic-regret magnitude carries genuine wake-model uncertainty. Reported numbers should be read as engineering estimates within this bracket rather than universal values.

4.4.2 Adversarial vs. realistic neighbors.

The greedy grid search places unconstrained individual turbines, while the cross-section uses a fixed reference farm copy. Neither represents a self-interested neighbor that optimizes its own AEP. A self-interested neighbor might concentrate turbines in high-wind upwind positions, potentially producing different regret patterns. This comparison is left to future work.

4.4.3 Wind speed treated as a per-sector constant.

Every result evaluates $P(\mathbb{E}[v])$ rather than $\mathbb{E}[P(v)]$: a single 9 m/s for the elliptical roses, and 24 sector-mean speeds for DEI. Because the power curve is strongly concave over its mid-range and flat above rated, collapsing the within-sector speed distribution onto its mean systematically over-states production. The effect is visible in the implied capacity factors—the DEI liberal layout reaches 5565 GWh on 750 MW, an 84.7% capacity factor, against roughly 50–55% for operating North Sea farms—so absolute AEP, the percentage denominators, and the monetary figures in Section 4 should be read as optimistic. Regret is a difference of two AEPs computed under the same treatment, so the comparative results are far less affected than the levels; a full speed-resolved re-evaluation of the headline cells is the obvious next step and requires no re-optimisation of the loss term.

4.4.4 Speed–direction correlation.

Real North Sea sites have speed–direction correlations. If high-speed winds are concentrated in the dominant direction (common for storm-driven sites), regret could be amplified through the cubic power relationship.

4.4.5 Single farm geometry.

Results may depend on the DEI polygon shape. Different boundaries could affect the layout optimization landscape and the potential for lateral escape from wake corridors.

4.4.6 Target farm size.

All main-text results fix $N_t = 50$ turbines. Larger farms have more internal degrees of freedom for re-optimization but also more internal wake to budget against. Appendix C reports a sweep over $N_t \in \{25, 75, 100\}$ at $\{2, 10, 20, 40\}D$ buffer for two contrasting wind roses; variation across N_t is at the scale of multistart noise.

4.4.7 Cross-section resolution.

The cross-section analysis uses 24 bearings and 7 distances, which may miss fine angular structure in the regret field.

5 Conclusions

1. **Reliable regret estimation requires many restarts and a strong inner optimizer.** With the tuned FunWake schedule, $K = 500$ captures a median 96 % of the $K = 2000$ regret when the signal is large (TurboPark) but only 54 % when it sits near the multistart noise floor (Bastankhah), and a weaker iteration schedule shifts the entire convergence curve downward. Any study computing AEP differences between strategies should use $K \geq 500$, store per-start AEPs, and verify convergence by subsampling.

2. **Design regret varies several-fold across the wind rose shape space, with folding (f) as the primary driver.** Realistic identical-copy neighbors at close spacing produce 0.19–0.49 % of AEP under Bastankhah and 2.0–10.6 % under TurboPark across the shape sweep; the concentrated unidirectional corner is the most vulnerable in both models.

3. **Buffer distance requirements are wind-rose- and wake-model-dependent.** Under Bastankhah, the fitted real North Sea sites require $\sim$8–$9D$ ($\sim$2 km) to hold regret below 0.2 % of AEP. Under TurboPark the same 2 % threshold is not reached at any of those sites within the $40D$ we swept, so the requirement is censored at $> 40D$ ($>$ 9.6 km)—already beyond the NYSERDA 4 nmi recommendation, with the true value unresolved. Wind-rose-conditional buffer standards could allow denser North Sea development at low-risk sites, but their absolute scale requires committing to a wake model.

4. **Adversarial upper bounds are only as good as the optimizer that computes them.** Under Bastankhah, the greedy individual-turbine bound (0.4–2.5 %) exceeds realistic identical-copy regret by roughly $5\times$. Under TurboPark, the realistic scenario computed with the stronger FunWake schedule (up to 10.6 %) overtakes the adversarial bound computed with the baseline optimizer (9.0 %)—under-optimization can masquerade as safety. Methodology validation with a degenerate single-direction case (Appendix A) confirms zero regret for downwind bearings, as required by physics.

5. **Multi-neighbor regret is non-monotonic, and bounded by lease geometry.** Regret decomposes into AEP loss times the fraction of it a redesign can recover. Adding neighbors raises the first and lowers the second, so regret peaks at intermediate ring sizes rather than growing: four close neighbors give only 1.2–$3\times$ the single-neighbor value, and beyond four, same-sized farms cannot pack close at all. Because a 50-turbine farm is itself $\sim 150D$ across, no more than four can ring a target anywhere in the 2–$40D$ buffer range; a fifth first fits at $42D$ (10 km), by which distance regret has collapsed. The many-close-neighbors corner of the parameter space is geometrically unbuildable rather than merely unstudied.

6. **What closes off redesign is concentrated wake mass, not symmetry.** Encircling a target lowers the recoverable fraction (≈0.6 to 0.38 under TurboPark), which invites a symmetry explanation. But dividing a *fixed* neighbor capacity into more, smaller farms at the same buffer spreads the threat around the compass and *raises* the recoverable fraction (0.51 to 0.65), because each individual wake corridor narrows. Concentration of wake mass in the directions that carry energy, not angular symmetry, governs how much of the damage a neighbor-aware layout can avoid—and it accounts for the wind-rose result as well.

The Danish Energy Island, with its partially bidirectional wind rose ($f \approx 0.38$), sits in the low-to-moderate regret region. Sites in the southern North Sea with strongly unidirectional southwesterly winds are substantially more vulnerable. As North Sea clustering intensifies toward the 300 GW target, incorporating wind rose directionality into buffer distance standards and coordination frameworks will become increasingly important for efficient maritime spatial planning.

. The analysis and figure-generation scripts are available at https://github.com/kilojoules/cluster-tradeoffs. The wake simulator used here, pixwake—a JAX implementation of the wake models in PyWake (Pedersen et al., 2019), providing automatic differentiation and GPU-batched multistart optimisation—is not yet public; a release is planned. Results were produced with an internal version whose behaviour is characterised in Section 2.2. The elliptical wind rose parameterization uses the edrose package (Hart, 2025), available at https://github.com/kilojoules/edrose.

. JQ designed the study, developed the simulation framework, performed all computations, and wrote the manuscript. SK, ES, and PM contributed to methodology and manuscript revision. PER supervised the research and contributed to methodology and manuscript revision.

. One author (PM) is employed by TotalEnergies, an offshore wind developer. The study was designed and executed independently and the results are reported without input from that affiliation.

. Computational resources were provided by the LUMI supercomputer, hosted by CSC – IT Center for Science (Finland) and the LUMI consortium, and by the DTU Computing Center (gbar).

Appendix A: Single-direction methodology validation

As a sanity check on the cross-section methodology, we ran a degenerate wind rose with all energy in a single sector (90°, wind from the east). Figure A1 shows AEP loss (left, no re-optimization) and design regret (right) for this case (baseline schedule, $K = 500$). Regret is precisely zero for all downwind bearings (225–315°), confirming that re-optimization cannot extract benefit from a neighbor that casts no wake over the target. Peak regret reaches 1.25 % of AEP at bearings flanking the inflow; the dip directly upwind reflects that the liberal layout is already pre-adapted to self-wake from that direction. The converged FunWake configuration reproduces this pattern under Bastankhah (peak 1.26 % at 135°), while under TurboPark the flank structure disappears and the peak sits directly upwind at 38 % of AEP (Section 3.5). The southeast quadrant retains non-trivial regret out to $40D$ (0.37 % at 135°/$40D$); Figure A2 shows the flow field for this configuration, with the SE neighbor's wakes drifting westward and grazing the target's southern row. At 135°/$40D$ the neighbor induces 0.44 % AEP loss, of which regret captures 0.37 % (re-optimization avoids ∼84 % of the damage).

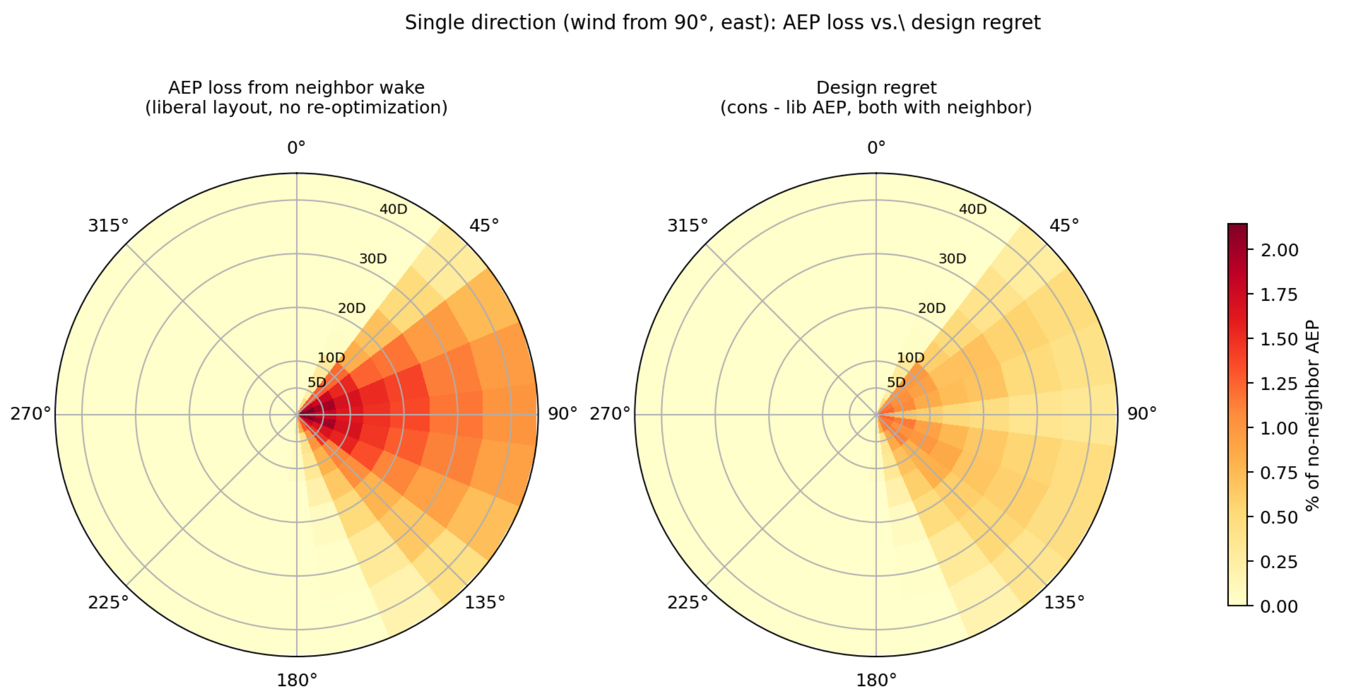

Figure A1. Single-direction validation case (wind exclusively from 90°): AEP loss (left) vs. design regret (right). Downwind bearings have zero regret as required by physics.

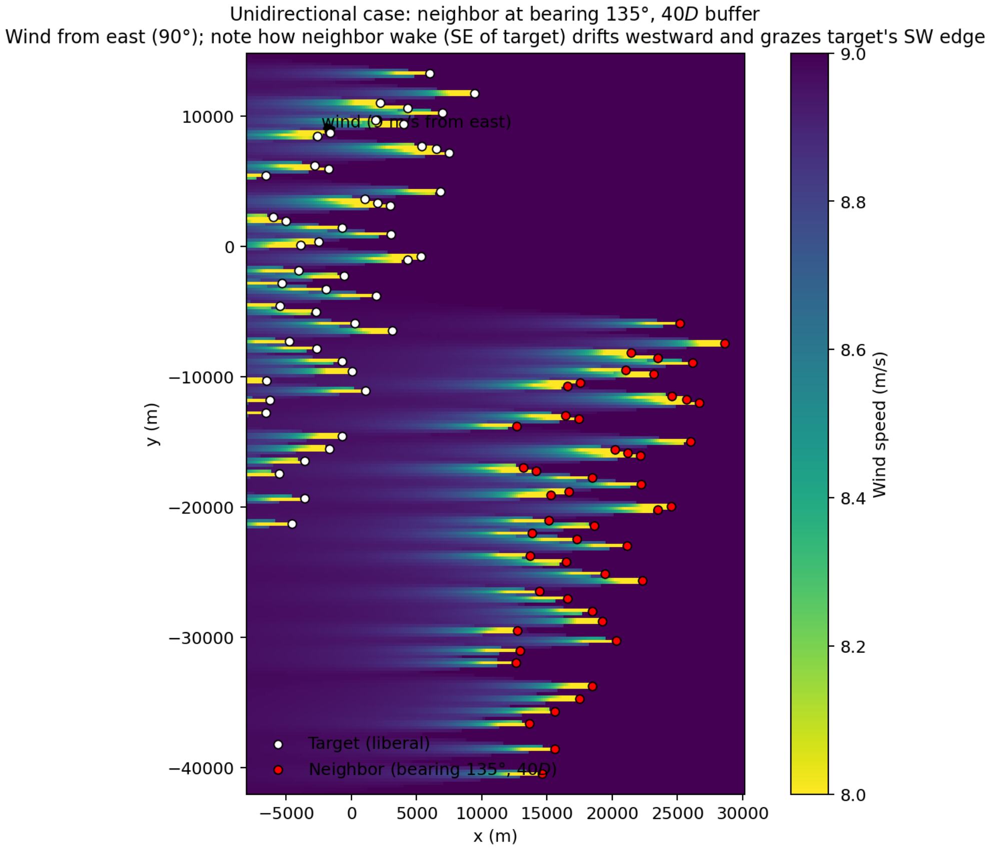

Figure A2. Flow field for $135^{\circ}/40D$ neighbor configuration, single wind direction (9 m/s from east). White circles: target liberal layout; red circles: identical-copy neighbor at bearing 135°, $40D$ boundary-gap.

Appendix B: Mixture wind roses (realistic multi-modal climates)

A single elliptical rose is a two-parameter caricature; real sites often have multiple directional modes (e.g. a dominant westerly plus a weaker easterly). We extend the analysis using Hart (2025)'s mixture parameterisation, which superimposes two ellipses at independent prevailing directions with a mixing weight. Eight representative configurations (Fig. B1) span orthogonal bimodal, asymmetric tri-modal, broad/narrow two-modes, North-Sea-like (SW+NE); this sweep uses the baseline schedule at $K = 500$. Peak regret across all mixture configurations stays within the single-rose envelope: at $d = 2D$ the worst mixture (asymmetric three-quarter, 0.41%) sits below the worst single rose ($a = 0.9, f = 1.0$; 0.47% in the same configuration), and decays similarly with buffer. The regret-maximising rose is not a multi-mode superposition—a single concentrated unidirectional rose is at least as bad.

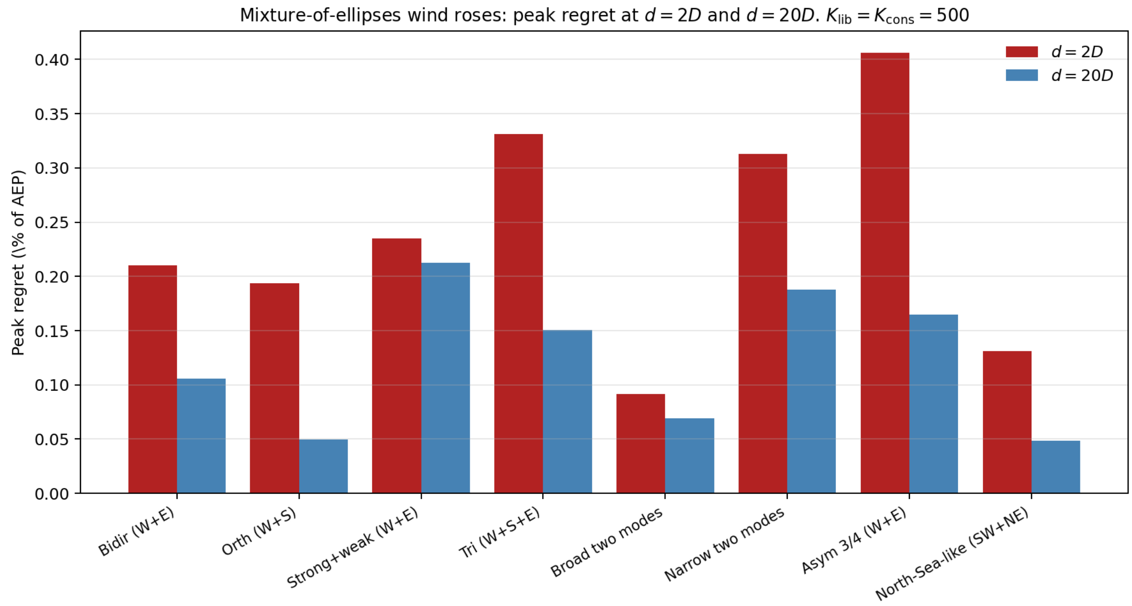

Figure B1. Peak regret for eight mixture-of-ellipses wind roses at $d = 2D$ (red) and $d = 20D$ (blue). All values stay within the single-rose envelope.

Appendix C: Target farm size sensitivity

We re-ran the cross-section at $N_t \in \{25, 50, 75, 100\}$ for the DEI rose and $(a, f) = (0.9, 1.0)$, at buffer distances $\{2, 10, 20, 40\}D$ with $K_{\text{lib}} = K_{\text{cons}} = 500$ (baseline schedule). The neighbor remains an identical translated copy of the target. Figure C1 shows peak regret vs. buffer distance for each N_t.

C0.1 Caveat: identical-copy is the wrong axis to upper-bound regret.

The above sweep keeps the neighbor as an identical copy of the target. This conflates target size with neighbor size, density, layout-correlation, and aspect ratio. A more rigorous upper-bound study would sweep the neighbor's construction (identical-copy, scaled-copy, independently-liberal-optimised in same / rotated / elongated polygon) at a range of size ratios N_n/N_t. We defer this two-dimensional sweep to follow-up work.

Appendix D: Greedy N-farm regret search

This is the greedy adversarial companion to the ring sweep of Section 3.6: instead of evenly-spaced rings, farms are placed sequentially wherever they maximise regret. For two wind roses, $(a, f) = (0.9, 1.0)$ and $(0.5, 0.0)$, we greedily placed up to $N = 4$ identical-copy farms maximising regret on the target (Bastankhah, baseline schedule, $K = 300$). Each step scans a 24×6 grid of (bearing, buffer-distance) candidates, filters by inter-farm boundary-gap $\geq 2D$, screens by AEP loss, and runs

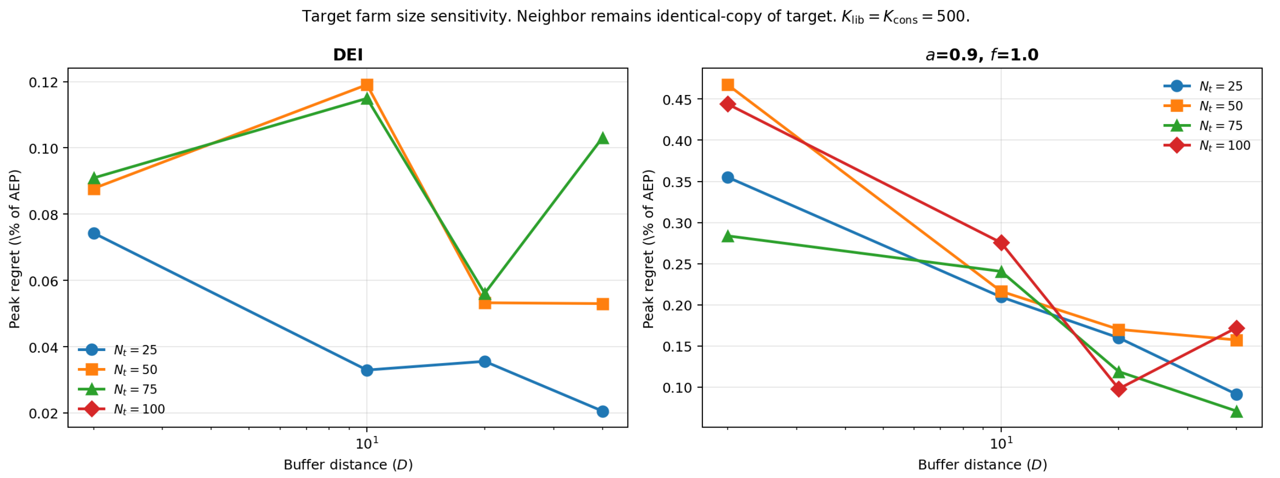

Figure C1. Peak regret vs. buffer distance at four target-farm sizes $N_t \in \{25, 50, 75, 100\}$. Neighbor is identical-copy throughout. Variation across N_t is at the same scale as multistart noise; no monotone trend in N_t is visible.

full conservative re-optimisation for the top 8 screened candidates. Table D1 reports the regret of the n-farm configuration after each greedy step.

Table D1. Regret (% of AEP) of the greedy n-farm configuration vs. number of placed farms (Bastankhah, $K = 300$).

Wind rose	$n = 1$	$n = 2$	$n = 3$	$n = 4$
$a = 0.9$, $f = 1.0$	0.42	0.53	0.56	0.56
$a = 0.5$, $f = 0.0$	0.68	1.12	1.02	1.03

In both cases, regret saturates well below $N\times$(single-neighbor regret): four adversarially-placed farms produce only 1.3–1.5$\times$ the single-farm regret (the two-farm configuration of the bidirectional rose peaks at 1.65$\times$), and for the bidirectional rose the third farm actually *reduces* regret relative to two. This matches the sublinear growth of the evenly-spaced ring; for why additional farms add so little recoverable damage, see the wake-mass argument of Section 3.6. Note that these absolute magnitudes are not directly comparable to the FunWake cross-section values in Table 4: here *both* the liberal and conservative optimizations use the weaker baseline configuration, and a weaker liberal layout leaves more recoverable damage on the table, which can raise regret even as a weaker conservative search lowers it. Only the saturation trend with N, which is internal to this experiment, should be read from this table.

Appendix E: The greedy search is not a maximiser over its candidate grid

The greedy grid search is described as establishing an adversarial ceiling, which presumes that enlarging the candidate set cannot lower the result: a maximiser over a superset should not score below a maximiser over a subset. We can test this directly, because two Bastankhah sweeps exist that differ *only* in grid extent— $50D$ of padding (1567 candidates) and $200D$ (10373 candidates)—at the same wake model, the same $K = 500$, and the same 30 placements.

The larger grid gives a *lower* regret in 33 of 35 shape cells, with a mean of $1.14\,\%$ against $1.48\,\%$ and a worst case of $1.80\,\%$ falling to $0.82\,\%$ $(2.2\times)$ at $a = 0.9$, $f = 0.0$. A superset cannot genuinely contain a worse maximum, so this is a property of the search rather than of the problem: the two-stage screen-then-evaluate scheme ranks candidates by the AEP loss they inflict on the *liberal* layout, only the top few advance to full evaluation, and the greedy path is sequential, so a changed candidate set diverts the whole trajectory. Enlarging the grid dilutes the screened shortlist and can steer the search down a worse path.

Two consequences for how the greedy numbers in this paper should be read. The Bastankhah sweep quoted in Section 3.2 uses the $50D$ grid and the TurboPark sweep uses $200D$, so the two are not merely different wake models but different search domains, and the cross-model ratio inherits that. More importantly, the greedy output is a *lower* bound on the worst achievable regret—the largest value any search happened to find—and calling it a ceiling is only defensible relative to the scenario class (individual unconstrained turbines rather than farms), not relative to the search. We report it on that basis throughout.

Appendix F: Sensitivity to the inter-neighbor spacing rule

The ring construction of Section 3.6 requires a minimum boundary gap between neighboring farms, set to $2D$ throughout. That constant decides when a ring is forced outward and therefore shapes the whole $n > 4$ region, so we swept it over $\{0.5, 2, 10, 20\}D$—a forty-fold range—under TurboPark at a requested $2D$ buffer, for both extreme wind roses (Fig. F1).

The rule changes the geometry it produces substantially: at $n = 4$ the realized target gap runs from $2.0D$ (at 0.5 and $2D$) to $4.7D$ ($10D$) and $9.2D$ ($20D$), a $4.6\times$ range. Regret magnitudes respond modestly, falling by about a tenth across that span ($12.2\,\%$ to $10.8\,\%$ at $n = 4$, concentrated unidirectional). The recoverable fraction—the quantity in which the mechanism of Section 3.6 is stated—moves by at most 0.044 and usually far less. Regret peaks at $n = 4$ in five of the six curves; the exception is the concentrated unidirectional rose at the $20D$ setting, where $n = 2$–4 lie within 0.3 percentage points of each other and the ordering within that plateau is not resolved. For $n \leq 3$ the neighbors are naturally 50–$70D$ apart, so the constraint never engages and all four settings are identical by construction. We therefore report the $2D$ results in the main text without qualification.

Appendix G: Wake-model robustness

The full cross-section and buffer-table analyses are replicated under the TurboPark deficit model (Nygaard et al., 2022) with ambient TI = 0.06, all other settings identical to the Bastankhah production runs. Table G1 compares peak regret on the DEI rose at the converged FunWake configuration.

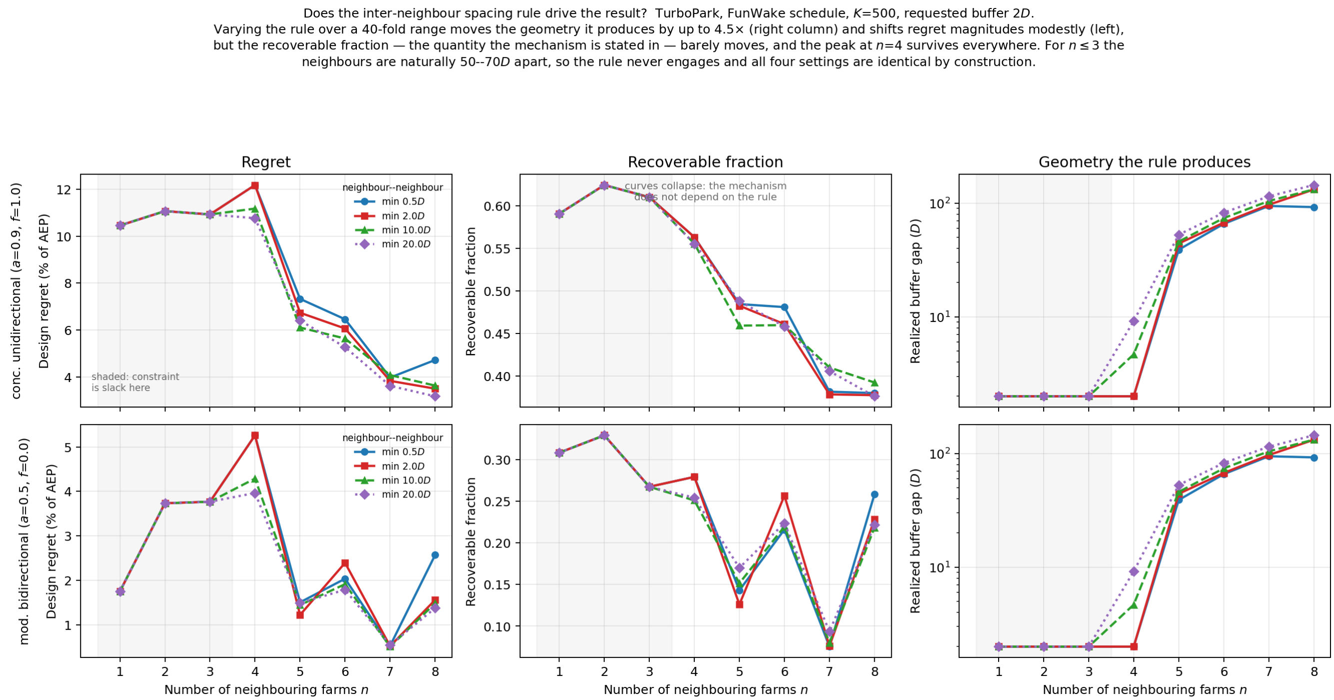

Figure F1. Sensitivity to the inter-neighbor minimum boundary gap (TurboPark, FunWake schedule, $K = 500$, requested buffer $2D$). Right column: the geometry each rule produces, which differs by up to $4.6\times$. Middle column: the recoverable fraction, on which the curves collapse. Shaded region: $n \leq 3$, where the rule is slack and all settings coincide.

Table G1. Wake-model sensitivity of cross-section peak regret on the DEI rose (FunWake schedule).

Buffer	Bastankhah ($k = 0.04$)	TurboPark ($A = 0.04$, TI = 0.06)	Ratio
$2D$	0.16 %	1.94 %	12.1×
$20D$	0.05 %	1.03 %	21.7×

The qualitative agreements—monotone decay with buffer distance, peaks at the closest buffer—are preserved, while absolute magnitudes disagree by an order of magnitude because TurboPark's wakes recover more slowly than Bastankhah's, so a neighbour at the same boundary gap couples more strongly to the target. Neither model is closer to ground truth; both are engineering parameterisations validated against partially-overlapping operational datasets, with different choices about wake recovery, blockage, and turbulence interaction. The two values should therefore be read as bracketing the realistic-regret magnitude under engineering wake-model uncertainty, not as competing best estimates. The methodological findings (catastrophic-cancellation multistart requirement, wind-rose-shape dominance, monotone buffer decay) are independent of this magnitude bracket.

G0.1 Loss-to-regret ratio.

The disparity in peak regret between the two wake models is best understood by comparing regret against the underlying AEP loss (liberal layout evaluated under the neighbour, no re-optimisation; Table G2):

Table G2. Peak-bearing AEP loss vs. design regret on DEI under both wake models (FunWake schedule). The ratio regret/loss is the fraction of neighbour-induced damage that target re-optimisation can recover.

Model	Buffer	AEP loss (% AEP)	Regret (% AEP)	Regret/loss
Bastankhah ($k = 0.04$)	$2D$	0.39	0.16	0.41
Bastankhah ($k = 0.04$)	$20D$	0.18	0.05	0.27
TurboPark ($A = 0.04$, TI = 0.06)	$2D$	4.67	1.94	0.42
TurboPark ($A = 0.04$, TI = 0.06)	$20D$	3.07	1.03	0.34

Three observations follow. First, regret is consistently 27–42$\%$ of AEP loss: target re-optimisation recovers only a fraction of neighbour-induced damage, because wake corridors cannot be fully escaped by layout shifts. Second, at the converged optimization the recoverable fraction at close spacing is essentially wake-model-independent (0.41 vs. 0.42 at $2D$). An earlier version of this analysis at a weaker optimizer configuration found a substantially lower TurboPark ratio (0.20–0.26), suggesting TurboPark damage was more "irreducible"; the converged results show that apparent irreducibility was an under-optimization artifact—the recoverable fraction is itself sensitive to optimization thoroughness, reinforcing the central methodological point of this paper. Third, one still cannot translate buffer requirements between wake models by a single multiplier: the regret ratio varies with buffer distance ($12\times$ at $2D$, $22\times$ at $20D$), so the regret-recoverable fraction must be computed model-by-model and distance-by-distance.

G0.2 Blockage sensitivity.

We additionally repeated the DEI cross-section with the SelfSimilarity2020 induction/blockage model added on top of Bastankhah (baseline schedule, $K = 500$). Peak regret rises modestly (0.09 % $\rightarrow$ 0.11 % at $2D$; 0.05 % $\rightarrow$ 0.10 % at $20D$ in that configuration): the upstream slowdown induced by a neighboring farm shifts the deficit field in a way that gives the conservative re-optimization slightly more room to extract benefit. The effect is at the same order as the optimization noise floor and does not change qualitative findings.

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